

Beyond the Spike at Zero: Understanding Nominal Wage Rigidity through Empirical and Model-Based Approaches*

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Abstract

I examine the cyclical properties of nominal wage change distributions across time and US states using nationally representative household surveys from 1979 to 2018. The novel finding is that the share of workers with no wage changes, which accounts for the large spike at zero in the nominal wage change distribution, is more countercyclical than the share of workers with wage cuts. I analyze heterogeneous agent models with six alternative wage-setting schemes and conclude that the model with downward nominal wage rigidity is the most consistent with empirical findings regarding the shape and cyclicalities of wage change distributions.

JEL Classification: E24, E32, J30.

Keywords: Downward nominal wage rigidity, nominal wage change distribution, wage stickiness, business cycle.

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1 Introduction

The sluggish adjustment of nominal variables plays a crucial role in resource allocation. Recent studies have theorized that downward nominal wage rigidity (DNWR) contributed to rising in unemployment in peripheral Europe and in the United States during the Great Recession ([Schmitt-Grohé and Uribe, 2016, 2017](#)). During periods of high inflation, real wages can fall even when nominal wages cannot adjust downward. In times of low inflation recession, during the Great Recession, however, it is believed that DNWR prevented real wages from falling, resulting in greater unemployment.

While most conventional models presume symmetric wage rigidity, asymmetric theories of wage rigidity can yield different allocative consequences for employment throughout the business cycle. For example, symmetric wage rigidity models fail to explain the skewness and the asymmetry observed in the unemployment rate – features that DNWR can explain ([Dupraz, Nakamura, and Steinsson \(2025\)](#)). Furthermore, the efficacy of macroeconomic stabilization policies aimed at boosting aggregate demand can vary based on the underlying wage rigidity model. For example, in a model with DNWR, the impact of government spending shock can be state-dependent ([Shen and Yang \(2018\)](#); [Jo and Zubairy \(2025\)](#)) or asymmetric ([Barnichon, Debortoli, and Matthes \(2021\)](#); [Born, D’Ascanio, Müller, and Pfeifer \(2024\)](#)), whereas in a model with symmetric wage rigidity, the effect remains linear and symmetric.

An open question remains: which theory of nominal wage rigidity best aligns with the cyclical movement of individual wage data? To shed light on this issue, I examine the cyclical properties of the nominal wage change distribution in relation to employment and inflation. I find that the share of workers with no wage changes exhibit more pronounced countercyclical patterns than the share of workers with wage cuts, both at the aggregate and state-levels. I then evaluate predictions from six different wage-setting models and compare them to these empirical patterns. The results suggest that DNWR most consistently aligns with the empirical findings.

This paper utilizes two nationally representative household surveys in the US: the Current Population Survey (CPS, 1979–2018) and the Survey of Income and Program Participation (SIPP, 1984–2013). Two key patterns emerge from the cross-sectional wage change distributions plotted using these data set: (i) a large spike at zero, and (ii) fewer wage cuts than raises.¹ This prominent asymmetry and the spike at zero in wage change

¹Figure [A2](#) and [A3](#) in Appendix [A.2](#) present the distribution for each year from 1979 to 2018 using the CPS. These graphs exhibit a consistent pattern across the years: asymmetric distribution with a spike at zero. Figure [A4](#) in Appendix [A.3](#) plots nominal hourly wage change distributions for hourly workers for each year from 1984 to 2013 using the SIPP data. All distributions exhibit asymmetry marked by a significant spike at zero, reflecting the patterns seen in the CPS-based distributions.

distributions have led many researchers to infer the presence of DNWR. However, relying solely on cross-sectional data is insufficient to confirm DNWR, as these patterns could also be influenced by factors such as labor productivity growth or inflation. Additionally, several studies challenge the DNWR, noting a significant proportion of workers experiencing wage cuts. Therefore, instead of focusing on cross-sectional wage change distributions, I examine the cyclical aspects of nominal wage change distributions.

My analysis focuses on three statistics that summarize the nominal wage change distribution between 1979 and 2018: the share of workers with no wage changes (which corresponds to the spike at zero), the share of workers with wage cuts and wage raises. This period captures significant fluctuations in inflation, making the interaction between inflation and nominal wage rigidity essential for understanding their real effects on employment. In theory, real wages, rather than nominal wages, determine employment levels. Hence, nominal wage rigidity would have minimal impact on employment during periods of high inflation recession, while it could adversely affect employment during periods of low inflation recession. Consistent with this prediction, I find that the spike at zero is not statistically significantly correlated with employment without controlling for inflation. However, the relationship becomes statistically significant when controlling for the inflation. This result supports the prediction that employment declines more sharply during a low inflation recession when nominal wage rigidity imposes stronger downward constraints, as reflected by a higher spike at zero.

Furthermore, I document a novel empirical finding—namely, that the spike at zero exhibits greater countercyclical fluctuations than the share of workers with wage cuts. As employment declines during recessions, fewer workers receive wage raises, while more workers experience either zero or negative wage changes. Among workers who do not receive wage raises, a larger proportion have no wage changes—resulting in a larger spike at zero—rather than experiencing wage cuts. Specifically, a one percentage point decline in employment is associated with a 0.9 percentage point decline in the share of workers receiving raises, a 0.6 percentage point increase in the spike at zero, and a 0.3 percentage point increase in the share of wage cuts, after controlling for inflation.

This pattern, identified at the national level over time, also holds in the cross-sectional analysis of data at the U.S. state level. State-level analysis offers two key advantages. First, it provides more observations, helping to alleviate issues related to limited degrees of freedom in aggregate analysis. Second, it enables state-level regressions with both state-and-time fixed effects, which control for state-specific trends and common factors across states, such as monetary policy or structural changes. Controlling for these fixed effects, declines in state-level employment continue to show a stronger correlation with

increases in the spike at zero than with increases in the share of wage cuts. Concerns may arise regarding the robustness of these results, given the prevalence of measurement errors in household surveys. To address this issue, I employ a structural model that explicitly incorporates measurement errors. The results indicate that measurement errors, in fact, dampen the excess cyclicalness of the spike at zero, further reinforcing the robustness of my findings.

In the subsequent section of the paper, I develop a model that incorporates existing wage rigidity settings and compare their predictions to identify which wage-setting mechanisms align with the observed cyclical patterns in the wage change distributions. This analysis builds on heterogeneous agent models featuring six distinct, but not mutually exclusive, wage-setting schemes widely discussed in the literature—perfectly flexible, Calvo, long-term contracts, symmetric menu cost, and two variants of DNWR models (an asymmetric menu cost model and an asymmetric Calvo model). Each model incorporates idiosyncratic labor productivity and aggregate shocks. The model assumes a complete market and is designed to be parsimonious, allowing for a quantitative solution to the heterogeneous agent framework while incorporating various wage constraints.

Using numerical methods, I first characterize the year-over-year wage change distributions implied by each model and examine how these distributions fluctuate with aggregate employment. I find that all models, except the perfectly flexible model, predict a stationary wage change distribution featuring a spike at zero. The time-dependent models—Calvo and long-term contracts—fail to generate the countercyclical movement of the spike at zero because they predict that the size of this spike remains constant over the business cycle. In contrast, the state-dependent models—symmetric menu cost and both DNWR models—generate a countercyclical spike at zero. However, the symmetric menu cost model predicts that, as employment declines, the share of wage cuts rises more than the spike at zero—a result inconsistent with the empirical findings.

Of all the models, only those incorporating DNWR is able to generate all key empirical patterns observed in the data. DNWR constrains downward wage adjustments, implying that the proportion of workers experiencing wage cuts exhibits little variation over time, whereas the spike at zero fluctuates more strongly throughout the business cycle. Moreover, the asymmetric Calvo model predicts the left-skewness of employment growth, a pattern consistent with empirical observations.

Combining empirical and theoretical findings, this paper introduces a new measure—the excess sensitivity of the spike at zero relative to the share of wage cuts—which can be used to assess the extent of DNWR not only in closed economies but also in open economies. While prior research has predominantly examined DNWR at the aggregate

level in the United States, relatively little attention has been given to state-level variation. Regional evidence is particularly valuable, as it can be interpreted through the lens of a monetary union with fixed exchange rates across states, as proposed by [Nakamura and Steinsson \(2014\)](#). Consequently, regional-level DNWR constraints can be viewed as analogous to the constraints faced by countries within a currency union.

For instance, [Jo and Oh \(2024\)](#) demonstrate that European Union countries with a more pronounced increase in the spike at zero relative to wage cuts experienced larger employment declines during the Great Recession, supporting the presence of DNWR. This finding is significant because it provides micro-data evidence of DNWR in these countries, helping to explain the massive increase in unemployment during the Great Recession, as highlighted by [Schmitt-Grohé and Uribe \(2016\)](#). Similarly, this approach can be extended to other open economies, for example, Ecuador, as shown by [Ahn, Choi, and Rivadeneyra \(2024\)](#). Their findings reveal that DNWR introduces significant frictions in propagating commodity shocks to labor market outcomes in small open economies.

State-level DNWR measures can also provide valuable insights into international shock propagation within the United States, after controlling for aggregate U.S. monetary policy. For example, [Rodríguez-Clare, Ulate, and Vasquez \(2024\)](#) examine how DNWR constraints amplify the effects of China shocks on the U.S. labor market. Using the state-level DNWR measures introduced in this paper, they show that states with higher DNWR experienced more severe effects from the China shock compared to states with lower DNWR.

Related Literature and Contribution In the empirical literature, there has been a long-standing debate over the existence of DNWR. [Kahn \(1997\)](#), [McLaughlin \(1994\)](#), [Gottschalk \(2005\)](#), and [Barattieri, Basu, and Gottschalk \(2014\)](#) underscore the asymmetric nominal wage change distribution. They argue that the relative scarcity of wage cuts compared to raises, along with the prominent spike at zero, provides evidence of DNWR. In contrast, other studies emphasize wage flexibility, highlighting the prevalence of reported wage cuts among workers: [Kurmann and McEntarfer \(2018\)](#); [Elsby and Solon \(2019\)](#); [Jardim, Solon, and Vigdor \(2019\)](#). Departing from these debates, which primarily focus on the shape of wage change distributions, my research investigates how these distributions respond to the business cycles.²

²The measurement of nominal wage rigidity can vary depending on factors such as data source, sample period, wage measures, and job types. Unlike this paper, [Fallick, Villar, and Wascher \(2022\)](#), [Kurmann and McEntarfer \(2018\)](#), and [Jardim, Solon, and Vigdor \(2019\)](#) use employer-reported earnings data to study wage rigidity. These studies report a lower share of workers with zero wage changes since they compute hourly wages as earnings divided by hours worked. In contrast, I focus on hourly wages for hourly workers, directly reported to the survey. A detailed comparison with earlier empirical literature on wage rigidity is

The empirical literature investigating the cyclical behavior of nominal wage rigidity has mostly focused on the cyclical spike at zero. [Card and Hyslop \(1996\)](#) report a negative correlation between this spike and inflation during the period 1979–1993, characterized by high inflation with an average rate of 5.3%. In contrast, [Daly and Hobijn \(2014\)](#) study the era of low inflation (1986–2014), with a 2.7% average inflation rate. They find that the spike at zero was countercyclical in low-inflation periods, increasing with declining employment. [Elsby, Shin, and Solon \(2016\)](#) challenge these findings, suggesting that the spike at zero has been acyclical since 1998 due to a secular decline in inflation. Building on these studies, my research investigates the cyclical behavior of the entire nominal wage change distributions—not just the spike at zero—in relation to both employment and inflation for an extended sample period from 1979 to 2018.

In related and contemporaneous work, [Grigsby, Hurst, and Yildirmaz \(2021\)](#) use payroll data from ADP, which offers the advantage of being free from measurement errors often associated with household survey data. They report evidence of both time- and state-dependent nominal wage adjustment from 2008 to 2016, which can be used for developing models of wage rigidity. My research, however, differs from theirs in several key aspects. I first expand the sample period from 1979 to 2018. This extended period, encompassing multiple business cycles and inflation levels, allows me to examine how nominal wage rigidity and inflation interact and their subsequent impact on employment. Moreover, in addition to presenting empirical findings, I utilize a heterogeneous agent model with various wage-setting schemes to identify which theory of wage rigidity best aligns with the observed cyclical properties of nominal wage change distributions.

My paper also relates to the theoretical literature. [Schmitt-Grohé and Uribe \(2016\)](#) propose a representative agent model featuring DNWR that limits the extent of aggregate nominal wage cuts. Subsequently, the model predicts a spike at a negative wage growth rate during recessions and no such spike during booms, indicating a binding DNWR constraint during economic downturns. [Fagan and Messina \(2009\)](#), [Daly and Hobijn \(2014\)](#), and [Mineyama \(2023\)](#) develop heterogeneous agent models with DNWR and characterize stationary wage change distributions that closely mirrors empirical wage change distributions. Building on this body of work, my theoretical analysis evaluates which wage rigidity models best match empirical findings, ultimately proposing that DNWR is most consistent with individual-wage data.

Notably, [Daly and Hobijn \(2014\)](#) and [Mineyama \(2023\)](#) argue that the countercyclical behavior of spike at zero can serve as evidence of DNWR. However, my analysis demonstrates that even a symmetric nominal wage rigidity model, such as menu cost model, can

available in Appendix [A.1](#).

also generate a countercyclical spike at zero. I therefore argue that it is crucial to examine the entire movement of the wage change distribution rather than focusing exclusively on changes in the spike at zero.

The remainder of the paper is organized as follows. Section 2 describes the data sets. Section 3 and Section 4 investigate the cyclical properties of the nominal wage change distributions in aggregate and state-levels, respectively. Section 5 builds heterogeneous agent models with six alternative wage-setting schemes. Section 6 evaluates numerical predictions from these six wage-setting schemes against the empirical findings. Finally, Section 7 concludes.

2 Data

This paper uses two nationally representative household panel data set in the United States, the CPS and the SIPP, with individual-level wage data. The primary subject of this paper is hourly workers who directly report hourly pay rates for both preceding year and the current year.³ For workers not compensated on an hourly basis, hourly wages can be computed by dividing the usual weekly earnings by the usual hours worked per week. However, this imputation method for determining hourly rates for salaried workers can lead to excessive volatile hourly wages due to its sensitivity to any reporting errors regarding the number of hours worked, known as the division bias (Borjas (1980)). To mitigate errors arising from the calculation of hourly pay rates for non-hourly workers, the main results are shown for hourly-rated workers. As of 2014, approximately 58 percent of workers in the United States were paid on an hourly basis.⁴ The main findings are robust to salaried workers as well.

This paper focuses on base wages for hourly workers, which excludes other types of benefits from total compensations, including paid leave, overtime payment, nonproduction bonuses, and so on. In December 2018, the BLS report on Employer Costs for Employee Compensation stated that 1.8 percent of total compensation could be attributed to

³In CPS, when households are in their fourth or eighth month of survey, they belong to the Outgoing Rotation Group (ORG). This group is asked additional questions related to earnings, such as their pay frequency, hourly wage, usual weeks worked, among others. Hence, the CPS can collect wage data twice a year, one year apart, from these households. While respondents are in the Outgoing Rotation Group (Month-In-Sample 4 or Month-In-Sample 8), they report their earnings in the easiest way: hourly, weekly, annually, or some other basis. Those who find it easiest to report their wage on an hourly basis are classified as hourly workers. However, there may be workers who do not initially report their earnings as hourly, but could be paid on an hourly basis. Consequently, those who stated that it is easiest to report their wages in a non-hourly manner are re-questioned if they are paid on an hourly basis, and if so, what their hourly pay rate is. The relevant questionnaire IDs are ERNPR, ERNHRTD, ERNRT, and ERNH2D, as per the Basic CPS Items Booklet.

⁴Source: <https://www.bls.gov/opub/reports/minimum-wage/archive/characteristics-of-minimum-wage-workers-2014.pdf>

nonproduction bonuses on average, while it was 1.4 percent of total compensation in December 2011. These findings suggest that nonproduction bonuses constitute a minor portion of the total compensation and do not exhibit cyclical properties. Similarly, [Grigsby, Hurst, and Yildirmaz \(2021\)](#) find that for median hourly workers, only about 2.7% of annual gross earnings come from compensation types other than base wages. They also argue that overtime and bonus compensation are quantitatively small and acyclical.

One limitation of the CPS is the difficulty in differentiating between job stayers and job switchers. While the CPS provides a variable to identify whether a respondent has the same employer as the previous month (available since 1994), data entry for this variable is missing in the fifth month of the survey after the eight-month break between interviews. This makes it challenging to accurately classify job stayers and job switchers using the CPS.⁵ Therefore, this paper does not distinguish job stayers from job switchers for the empirical analysis using the CPS.

Compared to the CPS, the SIPP offers a distinct advantage: the SIPP provides unique and consistent job IDs across waves for each job that the respondent had, whereas the CPS does not. Since job IDs are allocated based on a respondent's employer information in the SIPP, I define job stayers as employer stayers. Job switchers are those who reported to work for different employers in any given year, regardless of the jobless spell between employers. Despite this advantage, the SIPP data has certain limitations. First, the time series data is discontinuous due to gaps between the panels. Second, changes in survey methodology for collecting hourly wage data starting with the 2004 panel present additional challenges. In the initial interview, respondents were asked about their regular hourly pay rate. Beginning in 2004, however, respondents in subsequent interviews could indicate that their pay rate was "the same as the last interview," prompting the SIPP to automatically insert the previously reported rate. This methodological change complicates the consistent analysis of time series data using the SIPP.

This paper focuses primarily on hourly workers aged 16 and older. Self-employed workers and those whose earnings are top-coded or imputed are excluded from the analysis.⁶ The average number of observations in the CPS is 17,959 per year. For SIPP, the

⁵For instance, if a respondent switched jobs during the eight-month break period, for example, in the fifth calendar month, and remained in that job subsequently, they would report being employed by the same employer for the sixth to eighth survey months (MIS6-8). Note that the data entry indicating whether the respondent is employed by the same employer as the last month in MIS5 is missing. This respondent is likely to be incorrectly categorized as a job stayer from MIS4 to MIS8, even though they are, in fact, a job switcher.

⁶Wages, the most important variable in this paper, are often imputed in the CPS for missing values. On average, 34 percent of the hourly wages of hourly rated workers have been imputed since 1996. [Hirsch and Schumacher \(2004\)](#) show that including imputed wages in the analysis may cause bias due to the imperfect matching of donors with nonrespondents. Although IPUMS-CPS provides individually linked CPS data,

average number of observations is 13,511 per year, with approximately 71 percent classified as job stayers.⁷

3 The cyclicity of aggregate wage change distributions

In this section, I examine the aggregate-level cyclicity of nominal wage change distributions. To do so, I construct three statistics summarizing nominal wage change distributions: the share of workers receiving zero wage changes (the spike at zero), the share of workers with wage cuts, and the share of workers with wage raises, covering the period from 1979 to 2018. In order to understand how these statistics respond to the business cycle, I propose the following three regression equations:

$$\begin{aligned} [\text{Spike at zero}]_t &= \alpha_s + \beta_s(1 - e_t) + \epsilon_{st} \\ [\text{Share of wage cuts}]_t &= \alpha_n + \beta_n(1 - e_t) + \epsilon_{nt} \\ [\text{Share of wage raises}]_t &= \alpha_p + \beta_p(1 - e_t) + \epsilon_{pt} \end{aligned} \quad (1)$$

where e_t denotes the employment to population (epop) ratio in year t . If we sum up the three equations given above, we obtain: $1 = \alpha_s + \alpha_n + \alpha_p + (\beta_s + \beta_n + \beta_p)(1 - e_t) + \epsilon_{st} + \epsilon_{nt} + \epsilon_{pt}$, where the total on the left-hand side of equation (1) equals 1 because the three shares – spike at zero, share of wage cuts, and share of raises — form a complete probability density function. As the left-hand side of the equation is a constant, it can be inferred that $\beta_s + \beta_n + \beta_p = 0$. Consequently, β_p – the change in the share of workers with raises associated with the change in $(1 - e_t)$, can be decomposed into two parts: either β_s – the change in the spike at zero – or β_n – the change in the share of wage cuts.

Table 1 presents regression results based on the regression equation (1), both without and with controls for inflation. During periods of high inflation, nominal wage rigidity is likely to have a limited impact on real wage rigidity and, consequently, on employment. In contrast, during periods of low inflation, nominal wage rigidity may have a more pronounced effect on employment. Over the sample period from 1979 to 2018, inflation ranged from negative rates (e.g., -0.4 percent in 2009) to high rates (e.g., 12.7 percent in 1980). As a result, not controlling for inflation may understate the relationship between employment and nominal wage changes. Indeed, in the first three columns of Table 1, where inflation is not controlled for, I do not find statistically significant relationships between employment and the three statistics summarizing the wage change

the IPUMS-CPS does not provide allocation flags for wage variables, indicating whether wage variables are imputed or not. Therefore, I combine the IPUMS-CPS data with the monthly CPS, merged with the Outgoing Rotation Group, and exclude imputed wages.

⁷The time series number of observations is available from Table A3 in Appendix A.2. The time series number of observations is provided in Table A18, and the numbers of job stayers and job switchers are available in Table A19 in Appendix A.3.

Table 1: The spike at zero, the share of wage cuts, and raises along the business cycles

	(1) Spike at zero $\Delta W = 0$	(2) Share of cuts $\Delta W < 0$	(3) Share of raises $\Delta W > 0$	(4) Spike at zero $\Delta W = 0$	(5) Share of cuts $\Delta W < 0$	(6) Share of raises $\Delta W > 0$
1-Epop ratio ($1 - e_t$)	0.437 (0.560)	0.201 (0.442)	-0.638 (0.745)	0.617*** (0.217)	0.303 (0.258)	-0.919** (0.396)
Inflation rate (π_t)				-1.181*** (0.0606)	-0.673*** (0.158)	1.853*** (0.181)
Observations	38	38	38	38	38	38
Adjusted R^2	0.0442	-0.00342	0.0332	0.731	0.340	0.705

Data source: CPS and author's calculation. Sample Period: 1979-2018 (except 1995). The inflation rate is calculated from CPI-U. Heteroskedasticity and autocorrelation robust standard errors are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

distribution.

In contrast, when I control for inflation, as seen in the subsequent columns, I find statistically significant relationships. In particular, column (4) shows that the spike at zero increases when employment declines or inflation declines. This cyclical property of the spike at zero confirms the findings from the studies using shorter time periods by [Kahn \(1997\)](#), [Card and Hyslop \(1996\)](#), and [Daly and Hobijn \(2014\)](#). However, it challenges the argument presented by [Elsby, Shin, and Solon \(2016\)](#), which attributes the increase in the spike at zero to a secular trend rather than cyclical fluctuations.

Furthermore, the spike at zero exhibits greater countercyclical fluctuations compared to the share of wage cuts. I find that a 1 percentage point (pp) decline in employment is associated with (i) a 0.6 pp increase in the spike at zero, (ii) a 0.3 pp increase in the share of wage cuts—although this estimate is imprecise—and (iii) a 0.9 pp decrease in the share of raises. In other words, when employment declines by 1 pp, fewer workers receive wage raises (0.9 pp), and mechanically, more workers experience either no wage change or wage cuts (the sum of these two components must equal 0.9 pp). Notably, 67 percent ($= 0.6 / 0.9$) of this increase is attributable to the spike at zero, indicating that the rise in the spike at zero is substantially larger than the increase in the share of workers experiencing wage cuts.

Table 2 presents estimates from the regression of the spike at zero minus the share of wage cuts—a key measure in this paper—on the 1-epop ratio. The results indicate that the spike at zero is more responsive to the epop ratio than the share of workers with wage cuts, and this relationship is statistically significant. These findings are robust across both

Table 2: Excess sensitivity of the spike at zero relative to the share of wage cuts

	All			Prim-Age Men		Prime-Age Women	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Spike at zero minus share of wage cuts						
1-Epop ratio ($1 - e_t$)	0.237 (0.327)	0.314* (0.188)	0.424* (0.225)	0.487*** (0.182)	0.627*** (0.201)	0.197** (0.0990)	0.236** (0.117)
Inflation rate (π_t)		-0.508*** (0.146)	-0.471*** (0.125)	-0.320 (0.216)	-0.177 (0.157)	-0.540*** (0.164)	-0.529*** (0.162)
ΔTFP			0.551 (0.453)		0.780* (0.443)		0.562 (0.423)
Sample	All	All	All	Prime-Age Men	Prime-Age Men	Prime-Age Women	Prime-Age Women
Observations	38	38	38	38	38	38	38
Adjusted R^2	0.00	0.15	0.19	0.20	0.26	0.07	0.12

Data source: CPS and author's calculation. The inflation rate is calculated from CPI-U. Sample Period: 1979-2018 (except 1995). The first three columns show the results for all hourly-paid workers, columns 4 and 5 are for prime-age men, and columns 6 and 7 are for prime-age women. Heteroskedasticity and autocorrelation robust standard errors are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

prime-age (25–54) men and women, partially accounting for composition effects.

One may be concerned about measurement errors in self-reported hourly wages. Two main types of measurement errors can be expected with hourly wages. First, when respondents report their hourly wages, they may report their true wages with some errors. This type of measurement errors would understate the wage rigidity, the spike at zero. Second, workers may report rounded hourly wages, and this would overstate the spike at zero. Earlier studies that corrected measurement errors (Gottschalk (2005); Barattieri, Basu, and Gottschalk (2014); Murray (2019)) or used data free of measurement errors (Grigsby, Hurst, and Yildirmaz (2021)), argue that household typically report their wages with errors. This leads to a lower spike at zero in households survey data, thereby understating wage rigidity. For instance, Grigsby, Hurst, and Yildirmaz (2021) report that for hourly-paid job stayers between 2008-2016, the average share of workers with zero wage changes was 32.3%, which is much higher than the average size of spike at zero in the CPS, 15.2%.

In Section 6, I demonstrate that measurement errors reduce the excess sensitivity of the spike at zero, using structural models under the assumption that measurement errors are unlikely to vary with employment levels. This assumption is supported by the empirical finding that the fraction of imputed wages in the CPS, presented in the last column of Table A2, can serve as a proxy for the degree of measurement error. Notably, this

measure does not exhibit any cyclical variations. Building on this, I incorporate the distributions of measurement errors reported in the CPS by [Bound and Krueger \(1991\)](#) into the simulated wage data. I find that household survey data with measurement errors tend to underestimate overall wage rigidity and dampen the excess sensitivity of the spike at zero.

The primary results remain robust to salaried workers (non-hourly rated workers) and subgroups of workers by worker characteristics such as gender, age, race, and education. These robustness checks are detailed in Appendix [A.2.2](#). Furthermore, low-paid young workers, who are less likely to be employed under long-term contracts, also exhibit the main empirical findings. They display a sizable, and in fact, a greater spike at zero and a higher share of wage cuts compared to the overall sample. In addition, they demonstrate countercyclical fluctuations in both the spike at zero and the share of wage cuts, controlling for inflation, shown in Table [A15](#). This finding suggest that nominal wages are also rigid for those workers without long-term contracts.⁸

In order to explore the cyclical features of wage change distributions for job stayers and job switchers separately, I repeated the construction of the three statistics for each year using the SIPP data, categorizing the workforce into two groups: job stayers and job switchers. Table [A22](#) presents the regression results based on equation (1) using SIPP data. Job stayers show higher responsiveness of the spike at zero relative to the shares of wage cuts than job switchers. However, the wages of job switchers also exhibit countercyclical spike at zero. This phenomenon can be explained by [Krueger and Mueller \(2016\)](#), who suggest that the reservation wages for the unemployed are anchored to their previous wage. This result is also in line with [Hazell and Taska \(2024\)](#), showing downwardly rigid wages for new hires when unemployment rises.⁹

Notably, in the SIPP data, a significant increase in the spike at zero is observed from the 2004 panel onwards, attributed to changes in the survey methodology (see Table [A18](#)). To account for the different levels of the spike at zero, I incorporate a panel fixed effect

⁸I define low-paid young workers as hourly workers whose ages are less than 30 and hourly pay rates are less than the 25th percentile of hourly wages for each year and greater than the minimum wage. These workers constitute about 6 percent of the overall sample. For lower-paid young workers from 1979 to 2018, the average spike at zero and the share of workers experiencing wage cuts are 18.7 percent and 32.3 percent, respectively. Both of these figures exceed the averages for the overall sample, which are 15.2 percent and 21.1 percent, respectively. Low-paid young workers are less likely to experience rigid wages in a recession than the overall sample. Table [A15](#) suggests a 1 pp decline in employment is associated with (i) a 0.90 pp increase in the spike at zero; (ii) a 0.88 pp increase in the share with wage cuts, and (iii) a 1.78 pp decrease in the share with raises. In other words, 51 percent of workers having a zero wage among workers not having a raise.

⁹Supplemental materials on the number of observations and nominal wage change distributions using the SIPP data are available in Appendix [A.3](#).

into the regression, which is set to one for the years following 2004. The estimation is conducted with 24 observations with 3 regressors, which leaves the degree of freedom of 20. This lack of degree of freedom underscores the advantage of cross-sectional results presented in the next section, where time fixed effects control for changes in survey methods.

4 The cyclicalities of state-level wage change distributions

In this section, I investigate the cyclicalities of wage change distributions using the annual state-level panel data. State-level analysis offers two key advantages. First, state-level dataset provides me to exploit more observations, thereby mitigating any potential issues arising from the limited degrees of freedom in aggregate analysis. Furthermore, I can conduct state-level regressions incorporating both state-and-time fixed effects. State fixed effects account for state-specific characteristics, whereas time fixed effects control for factors common across states for each year, such as aggregate monetary policy, aggregate shocks, or structural changes.

4.1 State-level analysis from 1979 to 2018

I construct state-level annual share of receiving zero wage changes, wage cuts, and wage raises. To understand how state-level wage distribution responds to state-level employment, I propose the following state-level regression equations:

$$\begin{aligned} [\text{Spike at zero}]_{it} &= \alpha_{i,s} + \gamma_{t,s} + \beta_s(1 - e_{it}) + \epsilon_{it,s} \\ [\text{Share of wage cuts}]_{it} &= \alpha_{i,n} + \gamma_{t,n} + \beta_n(1 - e_{it}) + \epsilon_{it,n} , \\ [\text{Share of wage raises}]_{it} &= \alpha_{i,p} + \gamma_{t,p} + \beta_p(1 - e_{it}) + \epsilon_{it,p} \end{aligned} \quad (2)$$

where e_{it} is the employment to population (epop) ratio for each state i ($i = 1, \dots, 50$) and time t . α_i ($\alpha_{i,s}$, $\alpha_{i,n}$, and $\alpha_{i,p}$) capture state fixed effects, γ_t ($\gamma_{t,s}$, $\gamma_{t,n}$, and $\gamma_{t,p}$) absorb time fixed effects. As demonstrated in Section 3, controlling for state-level inflation is crucial.¹⁰ To account for state-specific differential time trends, I also incorporate controls for state-level female labor participation rate, male labor participation rate, the share of the population older than 65, the proportion of college-educated individuals, the share of service sector employee, and federal and state-level minimum wages.

Table 3 shows the regression results using the regression specification (2). A 1 percentage point decrease in employment is associated with (i) an increase in the spike at zero by 1.0 percentage points (Column (1)): (ii) an increase in the share of wage cuts by 0.6 per-

¹⁰State-level inflation data is from Nakamura and Steinsson (2014), Zidar (2019), and Regional Price Parities from BEA.

Table 3: The spike at zero, the share of wage cuts and raises across the US states

	(1) Spike at zero $\Delta W = 0$	(2) Share of $\Delta W < 0$	(3) Share of $\Delta W > 0$	(4) Spike at zero minus Share of wage cuts
1-Epop ratio ($1 - e_{it}$)	1.034*** (0.146)	0.593*** (0.118)	-1.627*** (0.225)	0.441*** (0.141)
State-level inflation (π_{it})	-0.174** (0.0729)	0.160* (0.0817)	0.0137 (0.105)	-0.333*** (0.114)
State Fixed Effect, α_i	Yes	Yes	Yes	Yes
Time Fixed Effect, γ_t	Yes	Yes	Yes	Yes
Control	Yes	Yes	Yes	Yes
Observations	1,823	1,823	1,823	1,823
Adjusted R^2	0.62	0.53	0.71	0.32

Data source: CPS and author's calculation. Sample Period: 1979-2018 (except 1985, 1986, 1995, and 1996 due to small sample sizes). The sample consists of 50 states over 35 years. State-level control variables include female and male labor force participation rate, the share of the population older than 65, the proportion of college-educated individuals, the share of service sector employee, and federal and state-level minimum wages. State-level cluster robust standard errors are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

centage points (Column (2)); and mechanically, (iii) a decrease in the share of wage raises by 1.6 percentage points (Column (3)). In other words, when employment declines by 1 percentage point, the share of wage raises also declines, and 63 percent ($=1.0/1.6$) of this change is attributed to the change in the spike at zero. The greater responsiveness of the spike at zero compared to the share of workers with wage cuts in the cross-section of US states (Column (4)) implies that state-level cyclical variations in nominal wage change distributions are consistent with the results from the aggregate analysis obtained in Section 3. In this context, each U.S. state can be viewed as analogous to a country in a monetary union with a fixed exchange rate regime. This framework can therefore be applied to test wage rigidity in open economies. Furthermore, these state-level measures can be used to examine how shocks propagate across states—within a monetary union with fixed exchange rates—depending on the degree of wage rigidity.

For the robustness check, I also perform regressions following specification in equation (1) for each state. The mean group estimator for the regression of the spike at zero on 1-epop ratio is 0.5, and for the regression of the spike at zero minus share of wage cuts on 1-epop ratio is 0.25. However, due to missing data over the sample period, several estimates are imprecise. When I condition on statistically significant estimates at 10 percent level, the mean group estimator for the regression of the spike at zero on 1 - epop ratio

is 0.71, and for the regression of the spike at zero minus share of wage cuts on 1-epop ratio, it is 0.6. The state-level regression estimates are available in Table A17 in online Appendix.

The findings in Table 3 encompass both job stayers and job switchers. To differentiate between the two groups, I execute the regression specifications (2) for each group separately using the SIPP data. The regression results are presented in Table A23. Note that time fixed effects account for common aggregate factors across states for each year, including the change in survey design that occurred in 2004. The pattern—greater countercyclicality of the spike at zero compared to the share of wage cuts—is present for both job stayers and job switchers, although the rigidity is much more pronounced for job stayers. These findings imply that the cyclical properties of nominal wage change distributions for all hourly workers are not solely driven by job stayers. This result is in line with Hazell and Taska (2024), showing downwardly rigid wages for new hires when unemployment rises.

4.2 Low-inflation recession: the Great Recession of 2007 - 2010

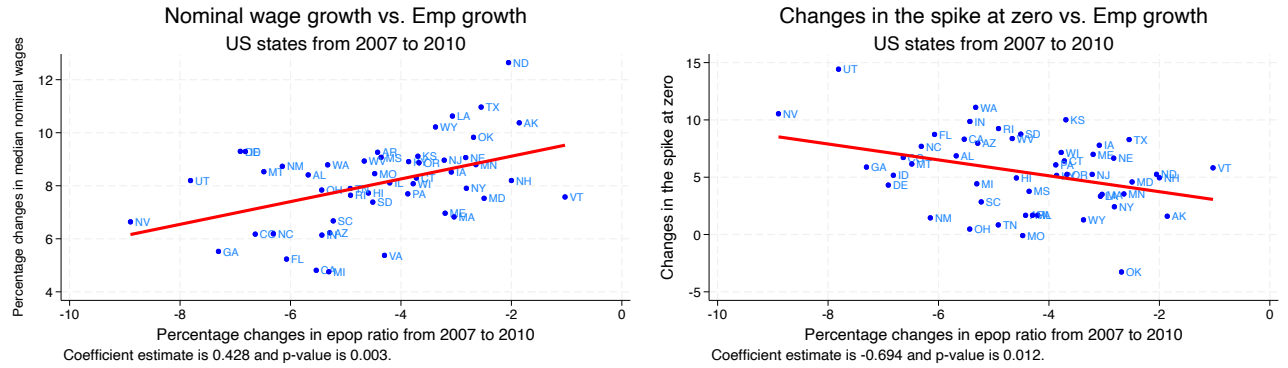
Departing from the findings documented in this paper, several studies have argued for greater wage flexibility using state-level wage data during the Great Recession. For example, Kurmann and McEntarfer (2018) suggest that the increased incidence of wage cuts during the recession indicates that DNWR may not act as a binding constraint during periods of large negative shocks. In another notable study, Beraja, Hurst, and Ospina (2019) (hereafter BHO) argue that there was only a modest degree of wage stickiness. One of their empirical findings is replicated in the left panel of Figure 1, which shows a positive correlation between nominal wage growth rates and employment growth rates across states from 2007 to 2010. Based on this relationship—where wage changes responded to employment changes—they estimate a relatively small wage rigidity parameter. As a result, they conclude that there was a fair degree of wage flexibility during this period.¹¹

The right panel of Figure 1 contrasts with BHO's approach by plotting the percentage changes in the spike at zero against the percentage changes in employment from 2007 to 2010 for each state. This plot reveals a negative correlation between changes in the

¹¹Using this state-level wage Phillips curve, BHO estimate the wage stickiness parameter through a partial wage adjustment model, in which the current wage is modeled as a weighted average of the optimal wage and the previous wage. They find that the weight on the previous wage is 0.31 to determine the current wage. Therefore, they conclude that there was only a modest amount of wage stickiness or a fair degree of wage flexibility.

The figure replicates Figure 1 of BHO using CPS data. A key difference between their study and mine lies in the construction of the wage data. BHO compute composition-adjusted average nominal wages for each state-year using the American Community Survey (ACS), which lacks a panel structure. Their sample consists of men between the ages of 21 and 55 with a strong attachment to the labor market only.

Figure 1: Nominal wage growth rates and changes in the spike at zero vs. employment growth across states from 2007 to 2010



Data source: CPS and author's calculation. The left panel plots the median nominal wage growth against employment growth rates from 2007 to 2010 across states. The right panel plots the changes in the spike at zero versus employment growth from 2007 to 2010 across states. From 2007 to 2010, the annualized inflation rate was 1.7 percent, and the cumulative inflation was 5 percent.

spike at zero and employment, showing that states with greater employment declines experienced a larger increase in the spike at zero.¹² In other words, states with larger declines in employment experienced more workers had rigid wages during more severe employment downturns.

Table 4 presents state-level regression results, corroborating these findings. While there was an increase in workers with wage cuts, as shown by Kurmann and McEntarfer (2018), there was an even more substantial increase in the spike at zero. This result is also compatible with BHO's empirical finding that shows a positive correlation between nominal wage growth rates and changes in employment (Column (4)). In states experiencing larger employment declines, a higher share of workers faced wage cuts, leading to an overall decline in nominal wage growth rates. However, these employment contractions are also accompanied by a substantially larger spike at zero, relative to the increase in the share of wage cuts. This suggests that a greater proportion of workers face wage rigidity in states undergoing more severe employment downturns. Consequently, the state-level data from the Great Recession continue to align with the patterns of wage rigidity documented in this paper, challenging the view that wages were fairly flexible or that DNWR may not have acted as a binding constraint during this period.

¹²One might be concerned about the small state-level sample size in the CPS data but the results are robust excluding small states such as Wyoming, Vermont, and Alaska. The coefficient estimate of the right panel graph of Figure 1 without those three state is -0.694 with a p-value of 0.022. Excluding outliers such as Utah, Vermont, and Oklahoma, the coefficient estimate is -0.462 with a p-value of 0.044.

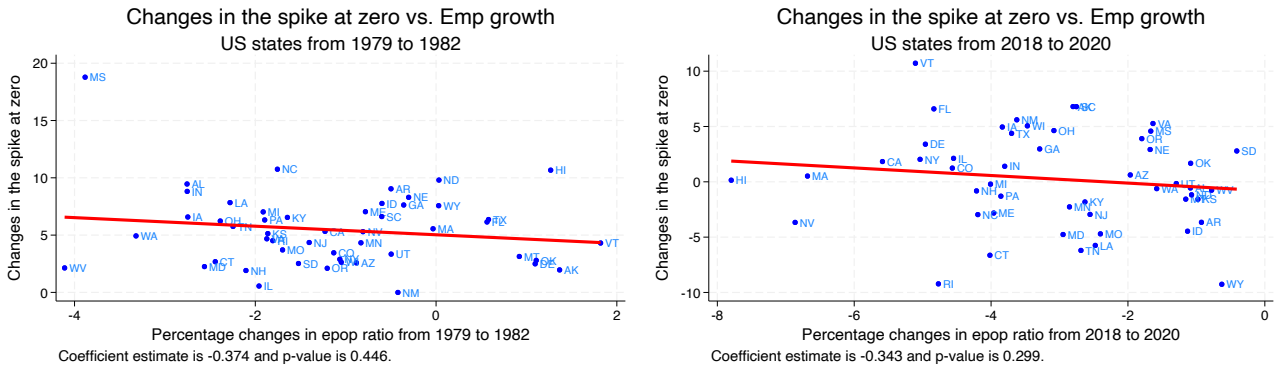
Table 4: Changes in nominal wage distribution from 2007 to 2010 across states

	(1)	(2)	(3)	(4)	(5)
	Changes in Spike at zero $\Delta W = 0$	Changes in Share of cuts $\Delta W < 0$	Changes in Share of raises $\Delta W > 0$	Changes in Spike at zero minus Share of cuts	$\ln \frac{W_{s2010}}{W_{s2008}}$
Percent change in employment rate	-0.694** (0.267)	-0.213 (0.320)	0.907** (0.397)	-0.480 (0.436)	0.428*** (0.136)
Observations	50	50	50	50	50
Adjusted R^2	0.104	-0.0104	0.0699	0.0103	0.183

Data source: CPS and author's calculation. Sample Period: 2007 - 2010. This table shows changes in nominal wage change distributions along with employment for each state from 2007 to 2010. Heteroskedasticity-robust standard errors are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

4.3 High-inflation recession

Figure 2: Changes in the spike at zero vs. employment growth 1979-1982 and 2018-2020



Data source: CPS and author's calculation. The left panel shows the change in the spike at zero with respect to employment growth from 1979 to 1982 across states. From 1979 to 1982, the average of annualized inflation rate was 9.5 percent and the cumulative inflation was 28.5 percent. The right panel shows the change in the spike at zero with respect to employment growth from 2018 to 2020 across states.

The Great Recession of 2007 - 2010 was a period of relatively low inflation. Thus, it was a period in which nominal rigidity resulted in real wage rigidity and, hence, real-locative effects on employment. One way to check whether nominal wages, as opposed to real wages, are rigid is to perform the same analysis just performed for the low-inflation recession of 2007 - 2010 for a high-inflation recession. In a high-inflation recession, nominal wage rigidity is expected to have a limited impact on employment, as real wages can adjust downwardly through inflation. In what follows, I consider the recession of

1979–1982 because it was a deep recession—comparable in size to the 2007–2010 recession—and characterized by high inflation, with the aggregate price level increasing by 29 percent between 1979 and 1982. Under the hypothesis that nominal wages, rather than real wages, are downwardly rigid, we should observe no significant relationship between employment changes and changes in the spike at zero across U.S. states during this period.

The left panel of Figure 2 plots changes in the spike at zero against employment growth rates across states from 1979 to 1982. In contrast to the Great Recession, no distinct relationship is observed between changes in the spike at zero and employment.¹³ Table A16 also confirms that the changes in nominal wage change distributions do not exhibit statistically significant relationships with employment. The right panel of Figure 2 illustrates the similar plots during the 2020 pandemic recession, which was followed by high inflation. Likewise, no discernible patterns emerge between the changes in the spike at zero and employment.

These findings indicate that during high-inflation periods, increases in the share of workers experiencing nominal wage rigidity—reflected in the change in the spike at zero—have limited impact on employment. This highlights the importance of controlling for inflation when examining the relationship between nominal wage rigidity and employment.

Furthermore, if real wages were rigid, one would expect the nominal wage change distributions to exhibit a spike around the inflation rate. However, this is contradicted by Figure A2. In 1980 and 1981, the spikes were observed at 7% (in the 6–8% bin) apart from the spike at zero, which is considerably lower than the inflation rates of 13% in 1980 and 10.4% in 1981. These results confirm that the rigidity we observe is about nominal wage rigidity, rather than real wage rigidity, highlighting nominal rigidity as the relevant friction in the labor market.

5 Heterogeneous agent models

This section develops a heterogeneous agent model with six distinct wage-setting schemes—though not mutually exclusive—including perfectly flexible wages, Calvo wage-setting, long-term contracts, symmetric menu costs, and two variations of DNWR. The model builds upon and extends the foundational work of Erceg, Henderson, and Levin (2000) and Benigno and Ricci (2011).

¹³Excluding outliers such as Mississippi, Hawaii, and West Virginia, the coefficient estimate is -0.271 with a p-value of 0.332. The results are robust in that there is no statistically significant relationship between the changes in the spike at zero and employment.

5.1 Firm

Consider a representative firm that produces consumption goods using the aggregate labor. The firm has a constant returns to scale production function in aggregate labor, $Y_t = L_t$, where Y_t is the output and L_t represents the aggregate labor. The profit function of the firm is $\Pi_t = P_t Y_t - W_t L_t$, where P_t is the price of goods and W_t is the aggregate nominal wage in the economy. There is no product price rigidity, and the firm's profit will be redistributed to the households. The firm's profit maximization problem is equivalent to the minimization of the cost of labor. Thus, the firm chooses differentiated labor $l_t(i)$, indexed by $i \in [0, 1]$, to minimize the total production cost, which can be written as

$$\min_{l_t(i)} \int W_t(i) l_t(i) di \quad (\text{s.t.}) \quad L_t = \int_0^1 (q_t(i) l_t(i))^{\frac{\theta-1}{\theta}} di,$$

given $W_t(i)$ is nominal wage for each individual i and $q_t(i)$ is idiosyncratic productivity for i . The aggregate labor L_t is a [Dixit and Stiglitz \(1977\)](#) aggregator of the efficiency adjusted differentiated labor. The problem of minimizing the cost of labor provides the labor demand function by the firm in the following form,

$$l_t^d(i) = q_t(i)^{\theta-1} \left(\frac{W_t(i)}{W_t} \right)^{-\theta} L_t, \quad \theta > 1, \quad (3)$$

where θ governs the elasticity of substitution across differentiated labor. The quantity of labor demand increases in the level of productivity and decreases in the relative wage. The aggregate wage W_t is defined as $[\int [W_t(i)/q_t(i)]^{1-\theta} di]^{1/(1-\theta)}$.

5.2 Households

There is a continuum of households, indexed by $i \in [0, 1]$, and each household chooses the consumption $C_t(i)$, saving $B_{t+1}(i)$, nominal wage $W_t(i)$, and labor supply $l_t(i)$ to maximize lifetime expected utility subject to intertemporal budget constraint, the labor demand function, and a wage-setting constraint. Households are assumed to have an additively separable preference between consumption and labor supply. Each household maximizes

$$\max_{\{C_t(i), B_{t+1}(i), W_t(i), l_t(i)\}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left[\frac{C_t(i)^{1-\gamma}}{1-\gamma} - \frac{\omega}{1+\psi} l_t(i)^{1+\psi} \right] \quad (4)$$

subject to the following constraints:

$$P_t C_t(i) + \mathbb{E}_t Q_{t+1} B_{t+1}(i) \leq B_t(i) + W_t(i) l_t(i) + \Pi_t,$$

$$l_t(i) = q_t(i)^{\theta-1} \left(\frac{W_t(i)}{W_t} \right)^{-\theta} L_t, \text{ and}$$

a wage setting constraint,

for given $\{P_t, Q_{t+1}, \Pi_t, B_0(i), L_t\}$. The wage setting constraint varies across the model, and the details will be given in the next section. Each household saves by $B_{t+1}(i)$ and Q_{t+1} represents the risk-free price of 1 unit of good for the next period. There are complete contingent asset markets so that idiosyncratic labor income is fully insured and the household consumes the exact same amount.

5.3 Six wage-setting constraints

As the household utility is additively separable, the labor supply decision can be isolated from the household's utility maximization (4), which can be re-written as

$$\max_{\{W_t(i), l_t(i)\}} \mathbb{E}_t \sum_{t=0}^{\infty} \beta^t \left\{ \lambda_t(i) W_t(i) l_t(i) - \omega \frac{l_t(i)^{1+\psi}}{1+\psi} \right\} \quad (5)$$

subject to the labor demand function (3) and a wage setting constraint. Here $\lambda_t(i)$ denotes the Lagrangian multiplier associated with the household's budget constraint in period t . This parsimonious specification allows me to quantitatively solve heterogeneous agent models featuring alternative theories of nominal wage rigidity. The first one is a perfectly flexible case, in which there is no wage-setting constraint. In this case, wage setters choose the optimal wage $W_t^{\text{flex}}(i)$ that solves the maximization problem (5) without a wage setting constraint for the entire period.

The second wage-setting scheme I explore is based on the Calvo rigidity model, which assumes that only a constant fraction of workers can optimize their wages in any given period. Followed by Calvo (1983), wage setters cannot optimize their wages with the constant probability of μ^{Calvo} , regardless of the state of the economy. The Calvo wage-setting restriction can be written as follows:

$$W_t(i) = \begin{cases} W_{t-1}(i) & , \text{ with the prob } \mu^{\text{Calvo}} \\ W_t^*(i) & , \text{ with the prob } (1 - \mu^{\text{Calvo}}) \end{cases},$$

where $W_t^*(i)$ represents the wage level, which solves the maximization problem (5) subject to a wage setting constraint for the entire period except in period t . In other words,

when wage setters get the opportunity to adjust their wage to $W_t^*(i)$ in period t , they are aware that they will encounter the wage-setting constraint in future periods.

In the third wage-setting scheme, I explore a long-term contract model. This model addresses that workers frequently engage in long-term contracts with firms. Therefore, the present discounted value of expected nominal wages over the contract duration plays a significant role in determining employment levels, rather than the remitted or observed wages at each point in time. This perspective is called Barro's critique (Barro (1977)) or the efficiency-wage theory. To address this critique, Basu and House (2016) introduced long-term contracts in a New Keynesian model in which firms pay the same nominal wages (remitted wages) over the contract. In this model, there are two distinct types of wages are introduced: allocative wages and remitted wages. The former determines the level of employment, while the latter represents the wages that the firm actually pays to the workers. Under this scheme, firms first calculate allocative wages under the perfectly flexible wage scenario and then determine the remitted wages, such that the present discounted value of remitted wages equals that of allocative wages over the contract term. Following Basu and House (2016), the remitted wages for each i type of labor, $x_t(i)$, can be determined as follows:

$$\mathbb{E}_t \left[\sum_{j=0}^{\infty} (\beta(1-s))^j \frac{\lambda_{t+j}}{\lambda_t} W_{t+j}^{\text{flex}}(i) \right] = \mathbb{E}_t \left[\sum_{j=0}^{\infty} (\beta(1-s))^j \frac{\lambda_{t+j}}{\lambda_t} x_t(i) \right],$$

where s is the probability of renewing the contract and $W_t^{\text{flex}}(i)$ denotes the wages under perfectly flexible wage setting.

Next, I examine the symmetric menu cost model of wage rigidity, motivated by the empirical evidence that changes in nominal wage change distribution are state-dependent. In this model, wage setters are required to pay a fixed cost, so called menu cost, to change their wages with a probability of μ^{Menu} . Conversely, with the remaining probability of $(1 - \mu^{\text{Menu}})$, wage setters can adjust their wages freely. This model echoes the random menu cost model in the price rigidity literature (Alvarez, Le Bihan, and Lippi (2016)), which explains small changes in prices within menu cost model. This wage setting constraint can be summarized as follows:

$$W_t(i) = \begin{cases} \begin{cases} W_t^*(i) & \text{if } W_t^*(i) \neq W_{t-1}(i), \text{ pays cost } K \\ W_{t-1}(i) & \text{no cost} \end{cases} & \text{,with the prob } \mu^{\text{Menu}} \\ W_t^*(i) & \text{no cost} \end{cases} \quad \text{,with the prob } (1-\mu^{\text{Menu}}).$$

Lastly, I turn to two schemes which incorporate DNWR constraint: the asymmetric

menu cost and asymmetric Calvo models. In the asymmetric menu cost model, wage setters are required to pay an additional fixed cost (K^D) with a probability of μ^{Menu} in order to reduce their wages, whereas no such costs are associated with wage increases. When wage setters find that the optimal wage is lower than the previous wage, they need to compare the benefits of lowering their wages to the optimal level (accounting for the fixed cost) with the benefits of maintaining wages at the previous level (without incurring additional costs). Alternatively, wage setters can adjust their wages without incurring any costs, with a probability of $(1 - \mu^{\text{Menu}})$. The wage-setting decision within the asymmetric menu cost model can be described as follows:

$$W_t(i) = \begin{cases} \begin{cases} W_t^*(i) & \text{if } W_t^*(i) \geq W_{t-1}(i), \text{ no cost} \\ W_t^*(i) & \text{if } W_t^*(i) < W_{t-1}(i), \text{ pays cost } K^d \\ W_{t-1}(i) & \text{if } W_t^*(i) < W_{t-1}(i), \text{ no cost} \end{cases} & , \text{with the prob } \mu^{\text{Menu}} \\ W_t^*(i) & , \text{with the prob } (1 - \mu^{\text{Menu}}) \end{cases} .$$

In the asymmetric Calvo wage setting, only a certain fraction of wage setters are able to lower their wages. If the optimal wage is higher than the previous level, the current wage can be adjusted to this optimal value - there are no explicit restrictions to raise the current nominal wage. However, if the optimal wage is lower than the previous level, then the wage setters cannot reduce wages with a constant probability of μ^{DNWR} . With the remaining probability of $(1 - \mu^{\text{DNWR}})$, wage setters can reduce wages to the optimal level. This wage-setting restriction can be summarized as follows:

$$\begin{aligned} & \text{if } W_t^*(i) \geq W_{t-1}(i) \left\{ W_t(i) = W_t^*(i) \right. \\ & \text{if } W_t^*(i) < W_{t-1}(i) \left\{ \begin{aligned} & W_t(i) = W_{t-1}(i) & , \text{with the prob } \mu^{\text{DNWR}} \\ & W_t(i) = W_t^*(i) & , \text{with the prob } (1 - \mu^{\text{DNWR}}) \end{aligned} \right. \end{aligned} .$$

These six different wage-setting schemes introduced in this section are widely used in the literature. For example, Calvo model of wage rigidity has been extensively applied in studies such as [Erceg, Henderson, and Levin \(2000\)](#); [Christiano, Eichenbaum, and Evans \(2005\)](#); [Smets and Wouters \(2007\)](#), among others. [Daly and Hobijn \(2014\)](#); [Aktuğ \(2021b\)](#) adopt the asymmetric Calvo model, while [Fagan and Messina \(2009\)](#); [Mineyama \(2023\)](#) use model with asymmetric menu costs. Notably, these six wage-setting schemes are not mutually exclusive. In fact, the asymmetric menu cost model, which includes more parameters than the other models, can encompass all models except the long-term contract

model.¹⁴

5.4 Closing the market

The goods market clearing condition is $Y_t = C_t$. Nominal output equals the total wage payment in the economy, which is the same as the total money supply in the economy.

$$P_t Y_t = P_t C_t = W_t L_t = M_t,$$

where M_t is the aggregate money supply. Monetary authority uses a nominal output growth rate targeting rule, given by

$$\ln(M_{t+1}) = \mu + \ln(M_t) + \eta_{t+1} \quad \eta_{t+1} \sim \mathbb{N}(0, \sigma_\eta^2), \quad (6)$$

where μ is the average growth of nominal output. The idiosyncratic productivity shock follows an AR(1) process as follows:

$$\ln(q_{t+1}(i)) = \rho_q \ln(q_t(i)) + \epsilon_{t+1}(i), \quad \epsilon_{t+1}(i) \sim \mathbb{N}(0, \sigma_\epsilon^2).$$

We can write down the household's wage-setting problem in a recursive way. Bellman equation for each model with six different wage setting schemes are available from Appendix A.4.3.

6 Numerical results

6.1 Solution methods

This section starts with a discussion of the numerical solution methods and the calibrated parameters. Table 5 reports the calibrated parameters. The top panel contains parameters relating to preference. The relative risk aversion parameter (γ) is set to 1, implying the intertemporal elasticity of substitution equal to 1. The discount rate (β) is 0.971, implying that the steady-state annual real interest rate is 3 percent. The inverse of Frisch elasticity (ψ) is set to be 0.5, which falls within the acceptable range in the macro literature shown in Chetty et al. (2011). The wage elasticity of labor demand (θ) varies widely in the existing

¹⁴In the asymmetric menu cost model, two distinct fixed costs are allowed: K^d for downward adjustment K^u for upward adjustment. To change wages, fixed costs have to be paid with a probability of μ^{Menu} , whereas it is free to change wage with a probability of $(1 - \mu^{\text{Menu}})$. The perfectly flexible case corresponds $\mu^{\text{Menu}} = K^u = K^d = 0$. Calvo wage rigidity can be represented by $K^d = K^u = \infty$ with $\mu^{\text{Menu}} = \mu^{\text{Calvo}}$. The symmetric menu cost model occurs when $K^d = K^u$. The asymmetric Calvo model is represented by $K^d = \infty$ while $K^u = 0$ with $\mu^{\text{Menu}} = \mu^{\text{DNWR}}$.

Table 5: Calibrated Parameters

Parameters	Value	Description	Target/Source
γ	1	Relative Risk Aversion	Annual interest rate, 3%
β	0.971	Discount rate	
ψ	0.5	Inverse of Frisch elasticity	
θ	3	Elasticity of substitution	
μ	0.044	Mean level of aggregate shock	Total wage payment
σ_m	0.021	Standard deviation of aggregate shock	Guvenen (2009)
ρ_q	0.821	Persistence of idiosyncratic shock	
σ_q	0.17	Standard deviation of idiosyncratic shock	
μ^{DNWR}	0.887	The probability of DNWR	
μ^{Calvo}	0.313	The frequency of no wage change	Grigsby et al. (2021)
s	0.323	The probability of continuing contract	
$\mu^{\text{Menu cost}}$	0.9	The probability of facing menu cost	
K	0.00254	Symmetric menu cost	
K^d	0.049	Asymmetric menu cost	

Note: Time unit is a year.

literature, ranging from 1.67 to 21.¹⁵ In line with Smets and Wouters (2007), I set θ at 3, implying steady-state markup 1.5.

The second panel of Table 5 shows the parameters governing shock processes in the economy. Since the nominal output is total wage payment in the model, I use total wage payment to estimate the aggregate shock process given by the equation (6).¹⁶ I estimate the constant growth rate and the standard deviation from the growth rate of the total wage payment. For the individual labor productivity shock, I use the stationary process of labor earnings from Guvenen (2009), allowing heterogeneity in the growth rate of income.¹⁷

The last panel of Table 5 presents parameters related to the degree of wage rigidity. As it is widely acknowledged that self-reported wage data, including that from the CPS, contain measurement errors, I calibrate the wage rigidity parameters to match with the size of the spike at zero as computed using ADP payroll data by Grigsby, Hurst, and Yildirmaz (2021), which are free from measurement error. Table 6 provides the average

¹⁵Erceg et al. (2000) set θ at 4. Christiano et al. (2005) set θ at 21. Smets and Wouters (2007) set wage markup at 1.5, which implies θ being 3. Daly and Hobijn (2014) set θ at 2.5. Fagan and Messina (2009) used $\theta = \frac{11}{12}$. Mineyama (2023) used θ at 9, which results in a steady-state wage markup of 12.5 percent. A recent paper by De Loecker and Eeckhout (2020) reported that the average markup in 1980 was 1.18, after which it began to rise and reached 1.67 in 2014.

¹⁶The total wage payment is defined as the median weekly earnings (Series ID: LEU0252881500) times the number of people at work (CPS series LNU02005053). Source: <https://www.bls.gov/data>

¹⁷Table 1, from Guvenen (2009). HIP (heterogeneity income process) after assuming $\sigma_\beta \neq 0$.

Table 6: Comparison of moments between empirical and simulated data

	Empirical data		Simulated data based on			
	Grigsby et al. (2021)	Calvo	Long-term Contracts	Symmetric Menu	DNWR	
Average					Asymmetric Menu	Asymmetric Calvo
Spike at zero	32.3	32.3	32.3	32.3	32.3	32.3
Share of cuts	1.7	13.7	11.9	13.3	3.9	3.7
Share of raises	66.0	54.0	55.8	54.4	63.8	63.9

Notes: Parameters governing the degree of wage rigidity are targeted to match with the size of spike at zero.

of the spike at zero, the share of wage cuts, and the share of wage raises using data from both ADP payroll data and model-generated data.

For a fair comparison between quantitative results with CPS data, this paper incorporates measurement errors into the simulated data. The distribution of measurement errors is taken from [Bound and Krueger \(1991\)](#), who compare matched CPS data with administrative Social Security payroll tax records and report the measurement error distribution. Detailed information about these distributions is provided in [Appendix A.4.1](#). It is also important to note that the models do not differentiate between job-stayers and job-switchers. Rather, the models aim to capture empirical findings based on evidence from the CPS, which includes both job stayers and job switchers.

I solve the recursive problem numerically using the policy function iteration over the discretized state space. The wage-setter's problem is infinite dimensional since they have to take into account the entire wage and productivity distribution. Following [Krusell and Smith \(1998\)](#), I assume that agents use only partial information, the first and second moments of the distribution, to predict the law of motion of the aggregate wage growth. For simulation exercises, I generate 5000 households for 5000 periods. Further details on solution methods are available in [Appendix A.4.4](#).

6.2 The cyclicalty of wage change distribution

This section examines whether the simulated wage change distributions resemble the empirical wage change distributions. I first evaluate how well the six models replicate the observed cross-sectional wage change distribution. [Figure A9](#) in the [Online Appendix A.4.2](#) illustrates the stationary nominal wage change distributions, accounting for measurement errors, generated by the six alternative wage-setting schemes. The stationary wage change distribution under the perfectly flexible case (top-left panel) displays a sym-

metric distribution around the median without a spike at zero. In contrast, all models apart from the perfectly flexible case display a spike at zero of 14.2%, as shown in red bar – a non-targeted moment. This spike is close to the CPS equivalent of 15.2%.¹⁸

The Calvo model (top-right panel) and the long-term contract model (second-left panel) generate a spike at zero and yield symmetric stationary wage change distributions around the median. These models fail to exhibit an asymmetric nominal wage distribution, which would be characterized by fewer wage cuts compared to wage raises. The symmetric menu cost model (second-right panel) results in a spike at zero and missing mass to the left of zero. Despite its symmetrical fixed costs for wage adjustments, it can also produce wage change distributions with a missing mass below zero. This occurs because the spike at zero includes workers whose desired wage adjustments are so small that they opt to maintain their wages rather than incur a fixed cost, regardless of whether the adjustment is upward or downward.

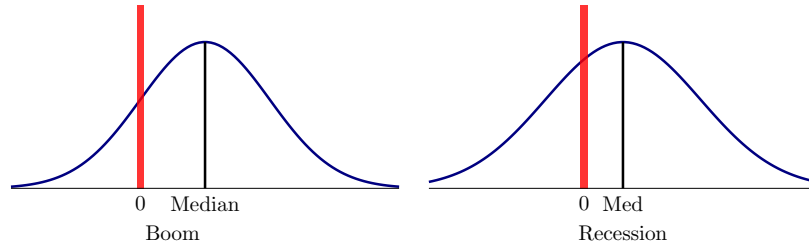
Both variants of the DNWR model (bottom panel) yield a spike at zero and a sharp drop below zero compared to above zero in the stationary nominal wage change distributions. These distributions exhibit asymmetry, with fewer wage cuts than raises and a sharp drop below zero compared to above zero. This result aligns with findings from [Fagan and Messina \(2009\)](#); [Daly and Hobijn \(2014\)](#); [Mineyama \(2023\)](#). However, the missing mass below zero can also be replicated by the symmetric menu cost model, making it difficult to attribute this feature solely to the presence of DNWR based on cross-sectional distributions alone. Therefore, I proceed to examine the cyclical aspects of nominal wage change distributions.

6.2.1 Intuition

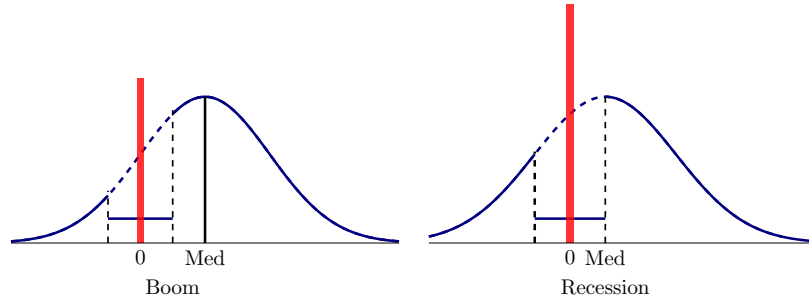
This section provides an intuition behind how each wage setting scheme implies cyclical-ity of wage change distribution. Figure 3 plots conceptual nominal wage change distributions along the business cycle under the alternative wage setting schemes. In addition, Table 7 reports the regression results of wage change distributions based on simulated data without measurement errors, using the equation (1), which regresses the spike at zero, the share of wage cuts and wage raises on employment and inflation.

Under a perfectly flexible wage model, the distribution of nominal wage changes shifts left or right along the business cycle. Panel A of Table 7 shows that, after controlling for inflation, a decrease in employment leads to an increase in the share of wage cuts that is nearly identical to the decrease in the share of wage raises. Additionally, the spike at zero does not react to changes in employment.

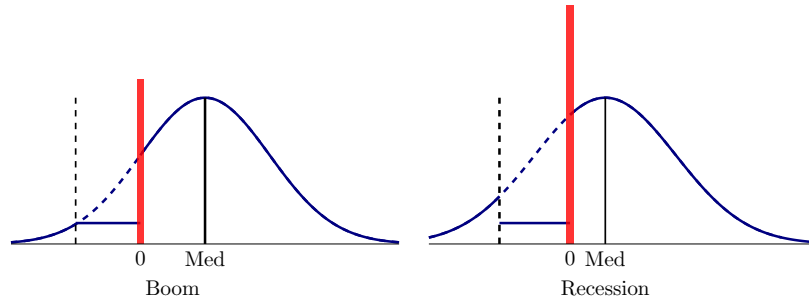
¹⁸Stationary nominal wage change distributions excluding measurement errors are shown in Figure A10.



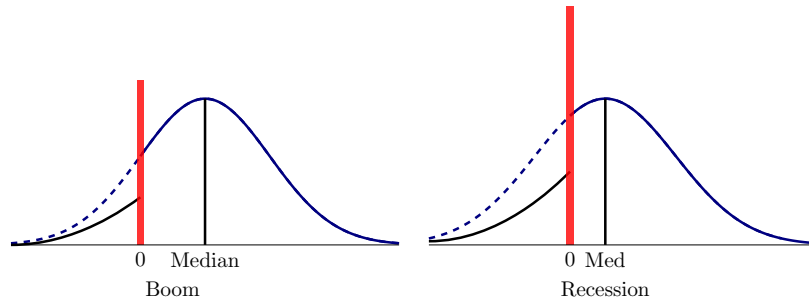
Calvo model



Symmetric menu cost model



DNWR with Asymmetric menu cost



DNWR with Asymmetric Calvo

Figure 3: Conceptual wage change distributions

This figure shows conceptual nominal wage change distributions along with the business cycles under symmetric wage rigidity constraints (Calvo and Symmetric menu cost model) and Asymmetric wage rigidity constraints (Asymmetric Menu cost and Calvo). In the absence of wage rigidity, the bell-shaped distribution shifts left in a recession compared to a boom. The red bar presents the spike at zero. The solid line represents the nominal wage change distributions in the presence of the corresponding wage setting scheme. The two dashed vertical lines in the Symmetric menu cost model represent the inaction region. The inaction region in the DNWR with Asymmetric menu cost model begins with the dashed vertical line below zero and ends at zero.

Under the Calvo model, the spike at zero remains the same throughout the business cycle. This is due to the fundamental assumption of the Calvo wage adjustment mechanism, where the frequency of no wage change does not respond to business cycles. The conceptual representation of changes in wage distributions under the Calvo model is depicted in the first panel of Figure 3. When employment declines, the distribution of wage changes shifts leftward, reducing the share of wage raises. This, in turn, leads to a corresponding increase in the share of wage cuts, while the spike at zero remains unaffected. This prediction of the Calvo model stands in contrast to the empirical finding of countercyclical spike at zero. These intuitions can be confirmed by Panel B of Table 7. Alternatively, we could imagine a variation of the Calvo model in which the frequency of wage adjustments is countercyclical. While this could generate the countercyclical spike at zero, it would not be able to produce an asymmetric wage distribution.

Under the long-term contracts model, similar to the Calvo model, the spike at zero remains constant throughout the business cycle (Panel C of Table 7). When employment declines, there is a decrease in the share of workers receiving wage raises, and this corresponds to an increase in the share of workers experiencing wage cuts of a similar magnitude. This finding again is not consistent with empirical finding of countercyclical spike at zero.

In the symmetric menu cost model, the spike at zero indeed responds to employment because this model is state-dependent. The second panel of Figure 3 illustrates the desired wage change distributions over the business cycle. During a recession, a negative aggregate shock shifts the nominal wage change distribution to the left. In the presence of symmetric menu costs, the model features a symmetric inaction region around zero in the wage change distribution, where wage setters refrain from adjusting wages due to fixed costs. This inaction region lies between the dashed vertical lines. With positive inflation, there is a higher density of desired wage changes near zero during recessions compared to booms. Consequently, a greater concentration of desired wage changes within the inaction region during recessions leads to a greater spike at zero. While the entire desired wage change distribution shifts to the left during a recession, only a fraction of workers in the inaction region keep their wages unchanged, contributing to increase in the spike at zero. This results in greater responsiveness of the share of wage cuts relative to the spike at zero. These predictions are supported by Panel D of Table 7. However, these predictions are inconsistent with empirical findings, which show greater responsiveness of the spike at zero than the share of wage cuts.

Previous studies often focus exclusively on the countercyclicity of the spike at zero as evidence supporting the presence of DNWR. However, the symmetric menu cost model,

which imposes symmetric frictions on adjusting wages both upward and downward, can also generate a countercyclical spike at zero. Therefore, relying solely on the countercyclical spike at zero as evidence of DNWR is insufficient. This discrepancy underscores the importance of analyzing the cyclical properties of the entire nominal wage change distribution rather than focusing exclusively on the spike at zero. Considering more comprehensively, the theoretical predictions of the symmetric menu cost model fail to align with empirical patterns documented across the entire distribution of wage changes.

The DNWR model predicts a countercyclical spike at zero, along with greater fluctuations in the spike at zero compared to the share of wage cuts (Panels E and F of Table 7). These predictions qualitatively align with empirical findings. During a recession, in the absence of wage rigidity, the nominal wage change distribution shifts to the left (third and fourth panels of Figure 3). In the asymmetric menu-cost model, a larger proportion of desired wage changes falls within the asymmetric inaction region (between the dashed line below zero and zero), which leads to a greater increase in the spike at zero relative to the rise in the share of wage cuts. Similarly, in the asymmetric Calvo model, 90 percent ($= \mu^{\text{DNWR}}$) of workers whose optimal wages fall below their previous levels experience no wage change, while the remaining 10 percent experience wage cuts. During a recession, a greater proportion of workers have optimal wages below their previous levels, resulting in a greater increase in the spike at zero than in the share of wage cuts.

6.2.2 Quantitative comparison between data and simulated model

To evaluate the models quantitatively, I compare the cyclical properties of wage change distributions using simulated data that incorporate measurement errors alongside CPS data. The objective is to identify which model most consistently aligns the cyclical patterns observed in wage change distributions. In summary, the key empirical findings are as follows: (i) the spike at zero increases when employment decreases, and (ii) the increase in the spike at zero exceeds the increase in the share of wage cuts. These findings are presented in Panel A of Table 8. To conduct this comparison, I first simulate wage data and incorporate measurement errors. Using these simulated data, I run regressions based on equation (1). Table 8 reports these regression results implied by each model: perfectly flexible (Panel B), Calvo (Panel C), Long-term contracts (Panel D), Symmetric menu cost (Panel E), Asymmetric menu cost (Panel F), and Asymmetric Calvo (Panel G).

Incorporating measurement errors into the simulated data dampens the cyclical properties of the wage change distribution. In both cases—with and without measurement errors—and while controlling for inflation, a decline in employment corresponds to a reduction in the share of wage raises. However, this decrease is less pronounced when measurement er-

Table 7: The model implied cyclicalities of wage distributions *without* measurement error

	Spike at zero $\Delta W = 0$	Share of cuts $\Delta W < 0$	Share of raises $\Delta W > 0$	Spike at zero $\Delta W = 0$	Share of cuts $\Delta W < 0$	Share of raises $\Delta W > 0$
	A. Perfectly flexible			B. Calvo		
Employment	-0.039 (0.065)	-0.414 (0.060)	0.454 (0.060)	0.095 (0.018)	-0.555 (0.028)	0.460 (0.029)
Inflation	-0.042 (0.025)	-4.476 (0.023)	4.519 (0.023)	-0.232 (0.018)	-3.509 (0.027)	3.741 (0.029)
	C. Long-term contracts			D. Symmetric menu costs		
Employment	0.008 (0.014)	-0.393 (0.035)	0.385 (0.035)	-0.248 (0.028)	-0.292 (0.032)	0.539 (0.029)
Inflation	-0.025 (0.009)	-3.810 (0.022)	3.835 (0.022)	-2.244 (0.011)	-2.907 (0.013)	5.152 (0.011)
	E. DNWR (Asymmetric menu costs)			F. DNWR (Asymmetric Calvo)		
Employment	-1.133 (0.040)	-0.199 (0.008)	1.332 (0.045)	-1.315 (0.040)	-0.163 (0.006)	1.479 (0.043)
Inflation	-5.130 (0.030)	-0.726 (0.006)	5.856 (0.033)	-5.163 (0.032)	-0.649 (0.005)	5.813 (0.035)

The table reports the regression results using simulated data series without measurement error under six alternative wage-setting schemes, based on the equation (1). Total 5000 periods of 5000 households are simulated. The numbers in parenthesis report standard error.

rors are taken into account. For instance, in response to a 1 percentage point (pp) drop in employment, the share of wage raises falls by 0.5 pp without measurement errors and by 0.3 pp with measurement errors under the perfectly flexible wage scheme. Similar patterns emerge across other models, with the share of wage raises decreasing from 0.5 pp to 0.4 pp in the Calvo model, from 0.4 pp to 0.3 pp in the long-term contract model, and from 0.5 pp to 0.4 pp in the symmetric menu cost model. The decline is more pronounced in models with asymmetric wage rigidities, falling from 1.3 pp to 0.8 pp in the asymmetric menu cost model and from 1.5 pp to 0.9 pp in the asymmetric Calvo model. These comparisons underscore the role of measurement errors in attenuating the cyclicalities of the wage change distribution.

The spike at zero in models with Calvo rigidity and long-term contracts remains acyclical, even after taking into account measurement errors. This outcome continues to diverge from empirical findings. While the symmetric menu cost model with measurement errors does yield a countercyclical spike at zero, it still suggests greater fluctuations in the share of workers with wage cuts than those with zero wage changes. This, too, is not consistent with empirical data.

Both DNWR models, even after incorporating measurement errors, exhibit (i) a coun-

Table 8: The model implied cyclicalities of wage distributions *with* measurement error

	Spike at zero $\Delta W = 0$	Share of cuts $\Delta W < 0$	Share of raises $\Delta W > 0$	Spike at zero $\Delta W = 0$	Share of cuts $\Delta W < 0$	Share of raises $\Delta W > 0$
<u>A. Data</u>						
Employment	-0.617 (0.217)	-0.303 (0.258)	0.919 (0.396)			
Inflation	-1.181 (0.0606)	-0.673 (0.158)	1.853 (0.181)			
<u>B. Perfectly flexible</u>						
Employment	-0.017 (0.029)	-0.327 (0.033)	0.343 (0.033)	0.040 (0.010)	-0.416 (0.020)	0.375 (0.020)
Inflation	-0.018 (0.011)	-3.401 (0.012)	3.420 (0.012)	-0.096 (0.010)	-2.800 (0.020)	2.897 (0.020)
<u>C. Calvo</u>						
<u>D. Long-term contracts</u>						
Employment	-0.002 (0.010)	-0.302 (0.026)	0.304 (0.026)	-0.098 (0.015)	-0.266 (0.023)	0.364 (0.023)
Inflation	-0.014 (0.007)	-3.027 (0.017)	3.040 (0.017)	-0.984 (0.006)	-2.682 (0.009)	3.666 (0.009)
<u>E. Symmetric menu costs</u>						
<u>F. DNWR (Asymmetric menu costs)</u>						
Employment	-0.500 (0.019)	-0.304 (0.013)	0.804 (0.026)	-0.581 (0.019)	-0.297 (0.012)	0.878 (0.025)
Inflation	-2.258 (0.014)	-1.633 (0.010)	3.891 (0.019)	-2.272 (0.015)	-1.590 (0.010)	3.862 (0.020)
<u>G. DNWR (Asymmetric Calvo)</u>						

The first row is from Table 1. Row 2, 3, and 4 show the regression results using simulated data series with measurement error under six alternative wage-setting schemes, based on the equation (1). Total 5000 periods of 5000 households are simulated. The numbers in parenthesis report standard error.

Table 9: The distance between empirical and model-implied cyclicalities wage distribution

L2 distance between empirical estimates and simulated data estimates on	DNWR					
	Perfectly flexible	Calvo	Long-term contracts	Symmetric menu costs	Asymmetric menu costs	Asymmetric Calvo
Employment	0.83	0.86	0.87	0.76	0.16	0.05
Inflation	3.35	2.61	2.88	2.71	2.50	2.46
Both	3.46	2.74	3.01	2.82	2.50	2.46

Notes: This table presents the square root of the squared distance (L2 distance) between the estimated coefficients reported in the last three columns of Table 1 and the coefficients estimated from the simulated data, as reported in Table 8.

tercyclical spike at zero and (ii) greater countercyclical fluctuations in the spike at zero compared to the share of wage cuts. The regression coefficients on employment are quantitatively similar to the empirical regression results from the CPS. Importantly, none of the parameters are calibrated to match the cyclicalities of wage change distributions or measurement error distributions. A comparison of regression results with and without measurement errors reveals that measurement error attenuates the excess sensitivity of the spike at zero relative to the share of workers with wage cuts.¹⁹

Moreover, both DNWR models qualitatively capture the excess sensitivity of the spike at zero to inflation relative to the share of wage cuts. However, the magnitude of coefficients on inflation is substantially larger than the empirical counterpart because the model assumes that price inflation equals wage inflation, abstract from price rigidity. Since the model does not factor in price rigidity, it projects more volatile responses to inflation than those observed in empirical data.

Table 9 summarizes the square root of the squared distance (L2 distance) between the regression estimates from Panel A of Table 8 and those derived from simulated data with measurement errors. Among the models, DNWR models – both the asymmetric Calvo and asymmetric menu cost – show the closest fit to the data for both employment and inflation regressors. This suggests that DNWR models provide the best explanation for the empirical evidence on the cyclicalities of wage change distributions.

In the symmetric menu cost model, two parameters, μ^{Menu} and the fixed cost, K , are calibrated to match the average spike at zero of 32.3%. Hence, these parameters are not uniquely determined. While maintaining the average spike near 32.3%, Table A25 reports that for varying values of μ^{Menu} and K , and across μ^{Menu} ranging from 0.4 to 1, the model consistently predicts a higher share of workers with wage cuts than the spike at zero. The square root of the squared distance between regression estimates on employment predicted by the symmetric menu cost model and the empirical data ranges from 0.76 to 0.78 – farther than the predictions made by DNWR models. The asymmetric menu cost models are also robust for μ^{Menu} greater than 0.9 in that showing greater fluctuations of the spike at zero compared to the share of wage cuts.²⁰

¹⁹This finding is qualitatively robust to an alternative parameter specification in which the Frisch elasticity is set to 1. That said, a higher Frisch elasticity leads to greater variation in employment compared to the baseline case but reduces the excess sensitivity of the spike at zero relative to the share of wage cuts. The results are presented in Table A24.

²⁰For μ^{Menu} less than 0.9, I find that no K less than 10 generates a spike at zero at 32.3%.

6.3 Implications for employment

One of the mechanisms through which monetary policy affects employment and output is the sluggish adjustment in nominal variables. In my model, the effect of monetary policy on aggregate employment—or the degree of monetary non-neutrality—can be quantified by the standard deviation of employment growth rates, similar to [Nakamura and Steinsson \(2010\)](#); [Midrigan \(2011\)](#). Table 10 contrasts data moments of wages and employment with the model implied moments by six alternative wage-setting schemes in response to the shocks to nominal spending. Note that, for computational tractability, the model considers only one aggregate shock to nominal spending, whereas the data moments may reflect the influence of multiple aggregate shocks.²¹

Panels B, C, and D of Table 10 compare moments generated by the symmetric wage adjustment models - Calvo, long-term contract, and symmetric menu cost model. The stationary nominal wage change distributions are summarized by three statistics: the average spike at zero, the share of wage cuts, and the share of wage raises. Note that the average spike at zero implied by model is about 14.2%, a non-target moment. This average spike at zero is close to the CPS equivalent of 15.2%. While the symmetric wage adjustment models yield similar wage change distributions, they have different implications on employment growth. The standard deviation of the employment growth predicted from the Calvo model (1.3) exceeds that predicted by either the long-term contracts (0.5) or the symmetric menu cost model (0.5).

The long-term contracts model leads to a smaller standard deviation of employment growth than the Calvo model because employment is determined by allocative wages, not remitted wages. The symmetric menu cost model also generates a smaller standard deviation of employment growth due to the selection effect, as suggested by [Caplin and Spulber \(1987\)](#) and [Golosov and Lucas \(2007\)](#). According to the menu cost model, only workers whose current wage levels significantly deviate from optimal levels wish to change their wages after incurring an additional fixed cost. After paying this fixed cost, these workers make substantial wage adjustments. This selection effect results in smaller allocative impacts on employment from aggregate uncertainty.

²¹For computational tractability, I use a shock to money stock as a representative example of an aggregate demand shock, which typically result in low-inflation recessions. In contrast, aggregate supply shocks—such as negative TFP shocks—generate different dynamics. These shocks lead to rising inflation alongside declining employment, where even nominal wage increases, if smaller than the increase in inflation, can result in downward adjustments in real wages. Consequently, the spike at zero does not increase significantly during high-inflation recessions. As a result, the relationship between employment and the nominal wage change distribution weakens during supply-driven business cycles, as illustrated in Section 4.3. Given these considerations and computational constraints, this paper focuses on shocks to the money stock as the primary source of business cycle fluctuations.

Table 10: Comparison between data and model generated moments

	Wage growth rate	Employment growth rate	Spike at zero $\Delta W = 0$	Share of cuts $\Delta W < 0$	Share of raises $\Delta W > 0$
<u>A. Data</u>					
Mean	4.228	-0.014	15.293	21.104	63.602
Std Dev	1.741	0.730	3.306	2.609	5.241
Skewness		-1.161			
<u>B. Calvo model</u>					
Mean	4.380	0.000	14.216	30.439	55.346
Std Dev	1.390	1.295	0.674	4.475	4.572
Skewness		0.044			
L2 distance from data			12.95		
<u>C. Long-term contracts model</u>					
Mean	4.364	0.004	14.218	29.492	56.290
Std Dev	1.215	0.495	0.495	3.877	3.889
Skewness		0.921			
L2 distance from data			11.69		
<u>D. Symmetric menu costs mode</u>					
Mean	4.376	0.000	14.225	29.681	56.094
Std Dev	2.058	0.537	2.160	5.644	7.642
Skewness		-0.021			
L2 distance from data			12.15		
<u>E. DNWR (Asymmetric menu cost) model</u>					
Mean	4.375	0.000	14.222	25.150	60.629
Std Dev	1.528	0.957	3.969	2.816	6.696
Skewness		-0.108			
L2 distance from data			5.39		
<u>F. DNWR (Asymmetric Calvo) model</u>					
Mean	4.376	0.000	14.222	25.251	60.526
Std Dev	1.496	0.965	4.017	2.707	6.630
Skewness		-0.118			
L2 distance from data			5.51		

Data source: CPS and author's calculation. Sample Period: 1979-2018 (except 1995). Wage growth rate is computed from the median hourly wage growth rate for hourly paid workers from 1979-2018. The model generated moments are calculated from the simulated data under six different wage setting schemes that incorporate measurement errors.

In Panels E and F of Table 10, I examine the moments generated by the asymmetric menu cost and the asymmetric Calvo models. These models maintain an average spike at zero similar to that of the symmetric rigidity models. However, due to the DNWR constraint, these models predict a lower share of wage cuts (25%) compared to their symmetric counterparts (30%).

The standard deviation of employment growth predicted by the DNWR models is 1. This value lies between the predictions of the symmetric Calvo model (1.3) and the symmetric menu cost model (0.5). These results suggest that, for a given degree of nominal wage rigidity—measured by the spike at zero—the DNWR models generate smaller real effects on employment than the symmetric Calvo model. The key reason for this difference is that DNWR imposes constraints only on wage reductions, leaving wage increases unconstrained. This asymmetric constraint distorts the labor market more during recessions than during expansions. Consequently, the smaller allocative impact on employment in expansions leads the DNWR models to predict lower overall standard deviations of employment growth compared to the symmetric Calvo model. Nonetheless, the predictions of the DNWR models remain larger than those of the symmetric menu cost model. This is due to the limited selection effect in downward wage adjustments.

To further assess the models' performance, I introduce an additional empirical moment—employment growth skewness. Employment growth rates are left-skewed, reflecting sharper declines in employment during recessions compared to increases during expansions. Panel A of Table 10 reports that the skewness of employment growth rates in the data is -1.161. Symmetric wage adjustment models, such as the Calvo, Long-term contracts, and symmetric menu cost models, lead to symmetric employment adjustments. As a result, these models imply skewness values of 0.044, 0.921, and -0.021, respectively.

In contrast, models incorporating DNWR capture the asymmetry in wage adjustments, which subsequently leads to asymmetric employment dynamics. During recessions, wage reductions are constrained, resulting in sharper employment declines. Conversely, during expansions, wages and employment rise, allowing the labor market to clear. Prior studies, including Dupraz, Nakamura, and Steinsson (2025); Aktuğ (2021a), highlight the role of DNWR in explaining the left-skewness of employment growth.

Among the two DNWR models examined in this paper, the asymmetric Calvo model accounts for a larger portion of the left-skewness in employment growth (-0.118) compared to the asymmetric menu cost model (-0.108). Although both models exhibit similar stationary and cyclical properties in wage change distributions, the average size of negative wage changes is significantly larger under the asymmetric menu cost model than under the asymmetric Calvo model. This difference stems from selection effects in the

asymmetric menu cost model. Firms most inclined to cut wages are willing to bear the fixed cost, leading to substantial wage reductions. Conditional on downward wage adjustments, the average size of wage change without measurement errors is -6.5% in the asymmetric menu cost model and -5.5% in the asymmetric Calvo model. This larger wage adjustment in the menu cost model reduces the skewness of employment growth relative to the Calvo model. As a result, the asymmetric Calvo model better captures the left-skewness observed in employment growth.

Finally, Table 10 reports the square root of squared distance between empirical moments and those predicted by the models. Among all the models considered, the DNWR models align closest with the empirical data. Within the DNWR models, the asymmetric menu cost model performs slightly outperforms the asymmetric Calvo model in matching empirical moments. The asymmetric Calvo model better captures the negative skewness of employment, although this advantage comes with trade-offs. Specifically, the asymmetric Calvo model predicts a lower standard deviation of wage growth rates (indicating more rigid wages) and a higher standard deviation of employment (indicating more substantial effects on employment) compared to the asymmetric menu cost model. In conclusion, while the asymmetric Calvo model may provide a better explanation for the skewed distribution of employment, it implies more severe allocative consequences for employment due to higher average wage rigidity.

Building on the conclusion that DNWR best describes the empirical micro wage data among widely used theories of wage rigidity, one might contemplate the distinct policy implications that DNWR offers compared to other wage rigidity models. This, however, is beyond the scope of the current paper. Instead, I summarize the findings from previous research on this question. [Benigno and Ricci \(2011\)](#); [Dupraz et al. \(2025\)](#); [Mineyama \(2022\)](#); [Aktuğ \(2021b\)](#) suggest that higher inflation can be welfare improving since it greases the wheels of the labor market, facilitating downward adjustments of real wages in the presence of DNWR. [Daly and Hobijn \(2014\)](#); [Mineyama \(2023\)](#) highlights the role of DNWR in flattening the Phillips curve during and after the Great Recession, as period characterized by low inflation recession. Similarly, [Schmitt-Grohé and Uribe \(2024\)](#) propose that DNWR can generate a non-linear Phillips curve. Additional work by [Shen and Yang \(2018\)](#); [Jo and Zubairy \(2025\)](#) show state-dependent government spending multipliers – an increase in government spending is more stimulative in a low inflation recession, when DNWR contributes to real wage rigidity. [Barnichon, Debortoli, and Matthes \(2021\)](#) present a model with DNWR can rationalize that the contractionary government spending multiplier is the largest in periods of economic slack. [Born, D’Ascanio, Müller, and Pfeifer \(2024\)](#) discuss the asymmetric effects of government spending - positive vs. nega-

tive - in a small open economy with a fixed exchange rate regime in a model with DNWR. Finally, [Ahn, Choi, and Rivadeneyra \(2024\)](#) show that DNWR can lead to severe labor market outcomes in response to negative commodity shocks in a small open economy with a fixed exchange rate.

DNWR may contribute to consumption inequality over the business cycle. However, the models in this paper assume complete markets. As a result, even when wages are constrained from adjusting downward, consumption remains perfectly insured. In fact, the model implies that households who work less due to the binding constraint may experience higher welfare, as they benefit from increased leisure. While computationally tractable, this is an unappealing feature of the model. Incorporating incomplete markets into the framework would address this limitation, but it poses significant computational challenges. While I do not expect incomplete markets to substantially alter the wage change distributions, I anticipate that they would lead to different outcomes in terms of consumption inequality. Empirical studies suggest that low-income households are more likely to face binding credit constraints, while low-wage workers are more likely to face binding DNWR constraints (Table [A14](#)). When both constraints are binding in response to large negative shocks, low-wage workers may experience significantly lower consumption, potentially increasing consumption inequality. However, this effect may be partially offset if low-wage workers respond by increasing labor supply in anticipation of such events. Investigating this topic further would be an interesting avenue for future research.

7 Conclusion

This paper investigates which model of nominal wage rigidity in the existing literature aligns most closely with micro-wage data in the US. To address this question, this paper first documents the cyclical properties of nominal wage change distributions using two nationally representative US household surveys: the CPS and the SIPP. The findings reveal that when employment declines after controlling for inflation, (i) the spike at zero increases; (ii) the share of workers with wage cuts also increases; and (iii) the increase in the spike at zero is much greater than the increase in the share of wage cuts.

Next, the paper develops a heterogeneous agent model with six alternative wage-setting schemes—perfectly flexible wages, Calvo, long-term contracts, symmetric menu cost, and two forms of DNWR—and examines their cyclical properties. The results show that models incorporating DNWR, in both the asymmetric menu cost or asymmetric Calvo models, most closely mirror the empirical patterns documented in the first part of the paper. This conclusion can suggest that DNWR may have allocative consequences

for employment.

Having established DNWR as a framework for describing the cyclical properties of micro wage data, future research could explore its broader macroeconomic implications within a richer model. For instance, incorporating price rigidity into the model would allow us to examine the interaction between nominal rigidities and their effects on output in response to monetary policy. Additionally, studying inequality arising from DNWR within an incomplete market framework—where idiosyncratic productivity shocks are not perfectly insurable—could provide further insights into distributional consequences.

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Online Appendix to “Beyond the Spike at Zero: Understanding Nominal Wage Rigidity through Empirical and Model-Based Approaches”

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This online appendix provides additional results and analyses. Section [A.1](#) offers comprehensive comparisons with previous literature. Section [A.2](#) contains the supplementary results from the CPS data, providing descriptive statistics and robustness checks for the primary analysis. Section [A.3](#) presents additional results from the SIPP data. Finally, Section [A.4](#) elaborates on further numerical methods and results.

A.1 Comparisons to the previous literature

The extent of wage rigidity is frequently gauged by the share of workers with zero wage changes. This measure of wage rigidity can vary depending on factors such as the sample period, data sets, pay type, and jobs. In this context, Table [A1](#) compares the average of the spike at zero, the proportion of workers receiving raises and wage cuts, thereby summarizing nominal wage change distributions in both my work and previous literature.

First, obtaining wages as earnings divided by hours yield a lower spike at zero and a higher proportion of workers with wage cuts. Wages calculated from earnings divided by hours are inherently volatile due to the larger measurement error associated with hours worked. This is known as division bias ([Borjas \(1980\)](#)). Second, job stayers exhibit a larger spike at zero compared to job switchers. This indicates a higher degree of wage rigidity among job stayers than job switchers, which is consistent with previous findings in the literature.

Third, the spike at zero can be underestimated due to measurement errors. Reported wages are often subject to measurement errors, which can distort the magnitude of the spike at zero. [Barattieri, Basu, and Gottschalk \(2014\)](#) use the SIPP panel data from the 1996 to 2000 period to estimate the constant frequency of no wage changes, after correcting for measurement error. They argue that this correction results in a larger estimate of the spike at zero and a decrease in the estimate of the share of workers with wage cuts. Similarly, [Grigsby et al. \(2021\)](#) utilize payroll data from ADP, free of measurement errors, and also observe a higher spike at zero and a lower share of workers with wage cuts.

These findings suggest that measurement errors in household surveys tend to underestimate the extent of wage rigidity.

Moreover, [Fallick, Villar, and Wascher \(2022\)](#) use data from the Employment Cost Index for the period from 1983 to 2019. This BLS survey provides information on the annual costs for specific job descriptions. The advantage of using employer-reported wage data is the absence of measurement errors. However, it does not allow for controlling for individual fixed effects since the base unit of observation is a job, not an individual. The authors conclude that the hypothesis that labor market distress during the Great Recession lowered nominal wage rigidity cannot be rejected.

[Kurmman and McEntarfer \(2018\)](#), using employer-reported earnings data from the Longitudinal Employer-Household Dynamics (LEHD) of Washington State, arguing that the increased incidence of wage cuts during downturns suggests that DNWR may not be a binding constraint. However, this paper reports even larger increases in the spike at zero compared to increases in the share of workers with wage cuts during downturns for the same sample period in Washington. [Jardim, Solon, and Vigdor \(2019\)](#) also use the same dataset from Washington to argue that nominal wage cuts are prevalent. They find a much lower spike at zero (4.4 percent) than my work (13.5 percent). This discrepancy arises because they calculate hourly rates as quarterly earnings divided by quarterly hours worked, rather than using base wages. Quarterly earnings include all monetary compensation such as base wage, overtime payment, and non-production bonuses. Consequently, their imputed hourly rates tend to be more volatile, resulting in a smaller share of workers with zero wage changes.

Numerous papers have examined nominal wage change distributions using the CPS. Figure A1 compares the spike at zero from my work and previous studies using the CPS. It is worth noting that when constructing the spike at zero, I include all hourly workers, both job stayers and job switchers, whereas previous literature typically only focuses on job stayers.

[Card and Hyslop \(1996\)](#) use the CPS from 1979 to 1993 to construct the share of workers with no wage change among hourly rated job stayers. [Elsby, Shin, and Solon \(2016\)](#) use the CPS from 1979 to 2012 and job tenure supplements to construct the share of workers with no wage change among hourly rated workers whose job tenure is over a year. The San Francisco Federal Reserve Bank has been publishing the Wage Rigidity Meter , which uses the CPS data starting from 1979 (with some gaps) and measures the share of workers with zero wage changes among workers who have not changed their jobs. ¹

¹For a fair comparison, I used the percentage of hourly rated job stayers with a wage change of zero from SF - Wage Rigidity Meter ([here](#)). In addition to hourly workers, non-hourly workers and all workers'

Table A1: Survey of the average spike at zero, the share of workers with raises and cuts

Paper	Data	Period	Wage measure	Pay type	Job stayers vs. Job switchers	Spike at zero $\Delta W = 0$	Fraction of $\Delta W < 0$	Fraction of $\Delta W > 0$
Kahn (1997)	PSID	1970-1988	Hourly pay rate Earnings/Hours	Hourly Salary	Job stayers Job stayers	9.8 4.8	10.3 23.4	79.9 71.8
Card and Hyslop (1996)	CPS	1979-1983	Hourly wage	Hourly	Job stayers	14.8	17.3	67.9
Elsby, Shin, and Solon (2016)	CPS CPS	1980-2012 1980-2012	Hourly wage Earnings/Hours	Hourly Nonhourly	Job stayers Job stayers	15.5 11.6	18.1 28.6	66.4 59.8
Wage rigidity meter (Daly and Hobijn (2014))	CPS CPS	1980-2018 1980-2018	Hourly wage Earnings/Hours	Hourly Nonhourly	Job stayers Job stayers	15.7 7.8	- -	- -
Jo (2025)	CPS CPS CPS	1979-2018 1979-2018 1979-2018	Hourly wage Earnings/Hours Earnings/Hours	Hourly Hourly Nonhourly	Both Both Both	15.3 10.0 6.9	21.1 27.4 34.2	63.6 62.7 58.8
Barattieri, Basu, and Gottschalk (2014)^a	SIPP SIPP	1996-2000 1996-2000	Hourly wage Hourly wage	Hourly Hourly	Job stayers Both	43.3 33.5	- -	- -
Jo (2025)	SIPP SIPP SIPP	1996-2000 1996-2000 1996-2000	Hourly wage Hourly wage Hourly wage	Hourly Hourly Hourly	Both Job stayers Job switchers	15.7 18.2 10.5	16.8 12.7 25.3	67.5 69.1 64.3
Fallick, Villar, and Wascher (2022)	ECI	1983-2019	Annual compensation/Annual hours	Hourly + Salary	Job stayers	15	16	69
Kurmman and McEntarfer (2018)	LEHD, Washington LEHD, Washington	2006 2010	Quarterly earnings/Quarterly hours Quarterly earnings/Quarterly hours	Hourly + Salary Hourly + Nonhourly	Job stayers Job stayers	7.5 ^b 16 ^b	20 ^c 25.0 ^c	72.5 ^d 59 ^d
Jo (2025)	CPS, Washington	2006	Earnings/Hours	Hourly + Nonhourly	Both	10.2 ^b	28.7 ^c	61.1 ^d
Jo (2025)	CPS, Washington	2010	Earnings/Hours	Hourly + Nonhourly	Both	13.1 ^b	37.8 ^c	49.1 ^d
Jardim, Solon, and Vigdor (2019)	LEHD, Washington	2005-2015	Quarterly earnings/Quarterly hours	Hourly + Nonhourly	Job stayers	4.4	24.8	70.8
Jo (2025)	CPS, Washington	2005-2015	Earnings/Hours	Hourly + Nonhourly	Both	8.1	32.7	59.1
Grigsby, Hurst, and Yildirmaz (2021)	ADP ADP ADP ADP	2008-2016 2008-2016 2008-2016 2008-2016	Hourly wage Base wage Base wage Base wage	Hourly Hourly + Salary Salary Hourly + Salary	Job stayers Job switchers Job stayers Both	32.9 5.2 34.8 28.6	1.8 38.0 3.6 8.7	65.3 56.8 61.6 62.7
Jo (2025)	CPS	2008-2016	Hourly wage	Hourly	Both	19.2	22.8	58.0

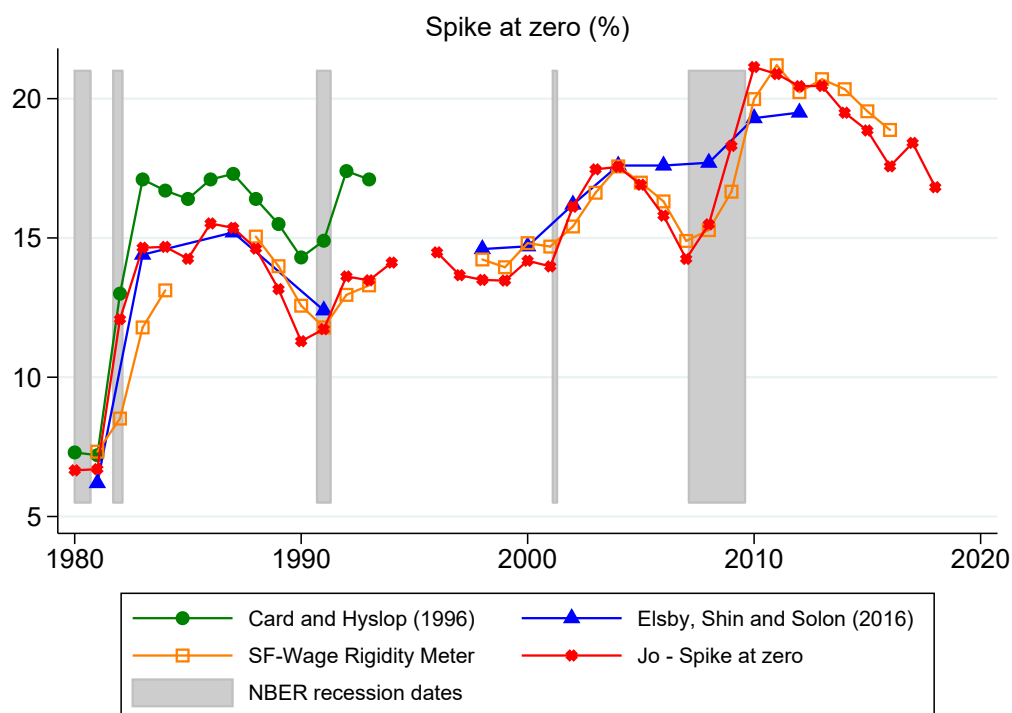
^a [Barattieri, Basu, and Gottschalk \(2014\)](#) report quarterly frequency of wage adjustment so it is converted to annual frequency.

^b Spike at zero contains log wage changes between -0.005 and 0.005.

^c The fraction of workers with wage cuts contains log wage changes less than -0.005.

^d The fraction of workers with raises includes log wage changes greater than 0.005.

Figure A1: Comparisons of the spike at zero from the previous literature



Notes:

Card and Hyslop (1996) - Data: CPS, Sample Period: 1979 - 1993, Job stayers only
 Elsby, Shin and Solon (2016) - Data: CPS, Sample Period: 1979 - 2012 (biannual), Job stayers only
 SF Wage Rigidity Meter - Data: CPS, Sample Period: 1979 - 2018, Job stayers only
 Jo (2023)- Data: CPS, Sample Period: 1979 - 2018, Both job stayers and job switchers

From the descriptions provided, one might expect the spike at zero as reported by [Card and Hyslop \(1996\)](#), [Elsby, Shin, and Solon \(2016\)](#), and the Wage Rigidity Meter to be similar. However, this is not the case. While these measures are highly correlated with each other, the actual levels of the spike at zero are different.

The spike at zero reported by [Card and Hyslop \(1996\)](#) is higher than that from [Elsby, Shin, and Solon \(2016\)](#) and the Wage Rigidity Meter. On the other hand, the spike at zero from [Elsby, Shin, and Solon \(2016\)](#) and the Wage Rigidity Meter aligns closely with the spike at zero that I have calculated, which includes both job stayers and job switchers from the CPS. However, when looking at the SIPP data, it appears that the spike at zero for job stayers is higher than that for job switchers. This might imply that the spike at zero as reported by [Elsby, Shin, and Solon \(2016\)](#) and the Wage Rigidity Meter does not come exclusively from job stayers.

A.2 Appendix: CPS

Table [A2](#) shows the unweighted number of population for age greater than 16 and the unweighted number of employed workers among the population greater than age 16. Table [A2](#) also reports the imputation ratio for usual weekly earnings and the hourly wage. After the major revision in the CPS in 1994, about 34 percent of hourly wages are imputed by the CPS. The CPS imputes unreported data entries to fill gaps based on the demographic characteristics and residential address.² This imputation adds additional measurement errors. [Hirsch and Schumacher \(2004\)](#) show that including imputed wages in the analysis can lead to bias due to the imperfect matching of donors with nonrespondents. As a result, this paper excludes imputed wages from the sample.

Table [A3](#) reports the number of observations for hourly workers whose hourly wage growth rate is available. Table [A3](#) also reports the time series of the national-level spike at zero and the share of hourly workers with wage cuts and raises.

(including both hourly and non-hourly workers) Wage Rigidity Meter is also available. The Atlanta Fed's Wage Growth Tracker ([here](#)) also reports the percentage of individuals with zero wage changes. However, their definition of individuals with zero wage changes includes those with hourly wage growth rates from -0.5 percent to 0.5 percent, while this paper and the SF - Wage Rigidity Meter only count workers with exact zero wage changes. Moreover, the Atlanta Fed's wage growth tracker includes both hourly workers and non-hourly workers, whereas this paper only considers hourly rated workers. They compute hourly wages for non-hourly workers by dividing usual weekly earnings by usual weekly hours worked or actual hours worked.

²<https://www.census.gov/programs-surveys/cps/technical-documentation/methodology/imputation-of-unreported-data-items.html>

Table A2: The unweighted number of observation in the CPS and the imputation ratio

Year	Age ≥ 16	Employed	Usual weekly earning			Hourly wage		
			Including Imputation	Excluding Imputation	Imputation ratio	Including Imputation	Excluding Imputation	Imputation ratio
1979	1,314,693	787,170	171,595	142,839	16.8	101,392	86,323	14.9
1980	1,546,827	918,046	199,290	167,183	16.1	116,941	100,699	13.9
1981	1,456,261	861,395	186,766	157,760	15.5	109,545	95,055	13.2
1982	1,404,030	813,120	175,643	151,075	14.0	102,475	90,129	12.0
1983	1,394,390	808,514	173,763	149,358	14.0	102,126	89,857	12.0
1984	1,374,456	819,764	176,724	150,317	14.9	104,287	90,780	13.0
1985	1,375,158	828,675	179,671	153,633	14.5	106,174	92,556	12.8
1986	1,353,321	821,067	178,586	159,172	10.9	105,861	96,029	9.3
1987	1,348,579	828,009	180,272	155,604	13.7	108,033	95,385	11.7
1988	1,286,466	797,107	172,931	147,658	14.6	104,079	90,836	12.7
1989	1,301,108	814,698	176,411	169,438	4.0	106,594	104,732	1.7
1990	1,355,294	846,099	185,030	177,254	4.2	110,923	109,005	1.7
1991	1,341,040	822,621	179,560	171,214	4.6	108,093	105,956	2.0
1992	1,320,939	808,261	176,848	169,030	4.4	107,005	105,270	1.6
1993	1,302,955	798,202	174,595	165,972	4.9	105,602	103,921	1.6
1994	1,271,347	790,130	160,156			104,717	82,639	21.1
1995	1,251,928	784,129	159,310	39,792	75.0	104,848	25,973	75.2
1996	1,108,899	699,605	141,174	109,580	22.4	93,894	71,033	24.3
1997	1,114,451	708,705	143,973	111,196	22.8	95,476	72,171	24.4
1998	1,116,813	717,245	145,834	111,960	23.2	95,918	71,122	25.9
1999	1,123,666	723,156	147,696	107,912	26.9	96,442	67,733	29.8
2000	1,120,585	723,930	150,104	105,873	29.5	97,234	65,844	32.3
2001	1,236,870	793,912	157,442	110,464	29.8	102,311	68,661	32.9
2002	1,312,304	832,519	171,206	119,583	30.2	110,671	74,040	33.1
2003	1,302,483	818,795	167,375	114,274	31.7	108,836	70,930	34.8
2004	1,283,683	809,185	164,286	112,821	31.3	107,382	70,234	34.6
2005	1,279,052	810,893	165,503	114,618	30.7	108,562	71,481	34.2
2006	1,271,693	810,582	165,888	114,382	31.0	107,510	70,495	34.4
2007	1,260,380	801,226	165,231	115,212	30.3	104,829	70,235	33.0
2008	1,257,619	790,341	163,450	113,582	30.5	102,940	68,390	33.6
2009	1,273,634	766,660	158,320	110,577	30.2	99,912	66,768	33.2
2010	1,277,199	759,458	156,751	104,806	33.1	99,512	63,764	35.9
2011	1,265,607	749,778	155,621	102,347	34.2	98,814	62,310	36.9
2012	1,258,730	749,477	155,207	103,279	33.5	98,263	62,453	36.4
2013	1,253,663	745,840	155,464	99,956	35.7	97,500	60,157	38.3
2014	1,261,811	751,675	156,924	98,855	37.0	98,227	59,129	39.8
2015	1,245,862	739,222	155,718	94,663	39.2	97,012	56,362	41.9
2016	1,244,166	740,071	156,389	95,943	38.7	97,466	57,354	41.2
2017	1,227,127	731,896	154,793	94,628	38.9	95,871	56,350	41.2
2018	1,188,950	710,991	151,367	91,414	39.6	93,619	54,552	41.7
2019	1,149,244	689,281	146,154	87,409	40.2	89,787	51,700	42.4
2020	1,034,141	581,093	126,778	78,215	38.3	74,011	44,275	40.2
Mean	1,273,034	778,632	163,948	123,191	25.6	101,921	75,540	26.4

Data source: CPS and author's calculation. Sample period: 1979 - 2018. This table shows the unweighted number of observation. The second column shows the unweighted number of individuals greater or equal to 16 for each year in the CPS. The third column shows the unweighted number of employed workers, greater or equal to age 16. Column 4-5 show the unweighted number of workers whose usual weekly earning is available including imputation (column 4), excluding imputation (column 5). Column 6 shows the imputation ratio for usual weekly earning. Column 7-8 show the unweighted number of workers whose hourly wages are available, including imputation (column 7), excluding imputation (column 8). Column 9 shows the imputation ratio for the hourly wage.

Table A3: The unweighted number of observation in the CPS and the imputation ratio

year	Unweighted count of		Spike at zero (%)		Share of cuts	Share of raises
	Δw	$\Delta w = 0$	Unweighted	Weighted	$\Delta W < 0$	$\Delta W > 0$
1980	21,029	1,403	6.67	6.66	14.24	79.11
1981	23,641	1,605	6.79	6.70	14.32	78.98
1982	23,211	2,839	12.23	12.08	18.90	69.01
1983	22,869	3,397	14.85	14.65	20.64	64.71
1984	22,840	3,398	14.88	14.68	20.21	65.11
1985	11,115	1,608	14.47	14.25	20.65	65.10
1986	6,202	956	15.41	15.52	21.48	63.00
1987	24,569	3,807	15.50	15.36	21.41	63.23
1988	23,302	3,414	14.65	14.62	20.38	65.01
1989	24,648	3,293	13.36	13.16	21.26	65.58
1990	28,898	3,282	11.36	11.29	23.44	65.27
1991	29,454	3,506	11.90	11.73	24.67	63.61
1992	29,247	4,009	13.71	13.62	25.37	61.01
1993	29,252	3,935	13.45	13.48	26.33	60.19
1994	22,971	3,253	14.16	14.11	23.89	62.00
1995						
1996	6,083	886	14.57	14.48	19.89	65.63
1997	18,057	2,533	14.03	13.66	19.56	66.78
1998	17,858	2,456	13.75	13.49	18.31	68.20
1999	16,876	2,346	13.90	13.47	18.95	67.58
2000	15,796	2,251	14.25	14.18	18.24	67.58
2001	14,719	2,062	14.01	13.98	18.64	67.38
2002	15,787	2,557	16.20	16.12	20.12	63.76
2003	17,333	2,932	16.92	17.46	21.08	61.46
2004	16,241	2,791	17.18	17.55	21.36	61.09
2005	14,991	2,466	16.45	16.91	20.63	62.46
2006	16,373	2,513	15.35	15.80	20.87	63.33
2007	16,245	2,309	14.21	14.25	20.43	65.32
2008	16,435	2,491	15.16	15.49	20.55	63.96
2009	16,069	2,906	18.08	18.30	23.59	58.10
2010	15,615	3,270	20.94	21.13	24.61	54.26
2011	14,774	3,030	20.51	20.88	24.30	54.81
2012	14,462	2,947	20.38	20.45	24.73	54.82
2013	14,463	2,896	20.02	20.46	23.07	56.48
2014	13,339	2,537	19.02	19.49	22.14	58.36
2015	10,752	1,975	18.37	18.86	21.58	59.56
2016	12,126	2,157	17.79	17.57	20.93	61.50
2017	12,674	2,322	18.32	18.41	20.26	61.33
2018	12,111	2,067	17.07	16.82	20.93	62.25
Mean	17,959	2,642	15.26	15.29	21.10	63.60

Data source: CPS and author's calculation. Sample period: 1979 - 2018 (except 1995). This table shows the number of observation and the spike at zero, the share of workers with wage cuts and raises for all hourly paid workers. Household identifiers were scrambles in 1995 so there were no observations available in 1995, and it leads to small observations in 1996.

A.2.1 Asymmetric nominal wage change distribution: 1980-2018

Figure A2 and Figure A3 illustrate the nominal year-to-year hourly wage change distribution for each year from 1980-2018. The nominal hourly wage change distribution is highly asymmetric: there is an apparent spike at zero and fewer wage cuts compared to raises. These characteristics are common to all nominal wage change distributions for each year for the entire sample period, 1979 - 2018.

Table A4 reports the averages of these three statistics across the sample years, 1980 - 2018. On average, 15 percent of hourly workers had exact zero hourly wage changes, 21 percent of them had wage cuts, and 64 percent had raises. Excluding minimum wage workers only has a marginal effect on these average estimates.³

Table A4: Descriptive statistics by worker characteristic, CPS

	% of all workers	% of hourly workers	Spike at zero $\Delta W = 0$	Share of cuts $\Delta W < 0$	Share of raises $\Delta W > 0$
Hourly Paid Workers	49.74		15.29	21.10	63.60
Exc. Minimum Wage Workers	45.17	90.89	15.15	20.62	64.22
Men	51.36	49.22	15.20	22.11	62.69
Women	48.64	50.78	15.38	20.08	64.54
16 ≤ Age <40	48.80	52.99	13.97	20.81	65.22
40 ≤ Age <64	47.61	43.05	16.01	21.65	62.34
Prime-Aged Men		46.27	16.30	21.69	62.01
Prime-Aged Women		49.91	16.54	20.80	62.66
High School Or Less	44.35	58.09	15.80	21.45	62.75
College Or More	55.65	41.91	14.52	20.65	64.83
White	85.74	85.02	15.41	20.56	64.03
Non-White	14.26	14.98	14.66	24.30	61.04

Data source: CPS and author's calculation. Sample Period: 1979-2018 (except 1995). This table shows the sample average of spike at zero and the share of workers with wage cuts and raises over time by worker characteristics.

Nominal hourly wage change distributions do not exhibit significant heterogeneity by gender, age, race, and education. Table A4 reports descriptive statistics by those worker characteristics. As this study only considers hourly workers, there's an overrepresentation of female workers, younger workers, and less educated workers. However, when we calculate the averages of the three statistics for different subgroups of workers, the estimates remain relatively consistent.

³Workers whose hourly wages are lower than the state's minimum wage in either the previous or current year are dropped. Data source: Vaghul and Zipperer (2016); <https://www.dol.gov/whd/state/stateMinWageHis.htm>.

Hourly paid workers, CPS, 1980-2000

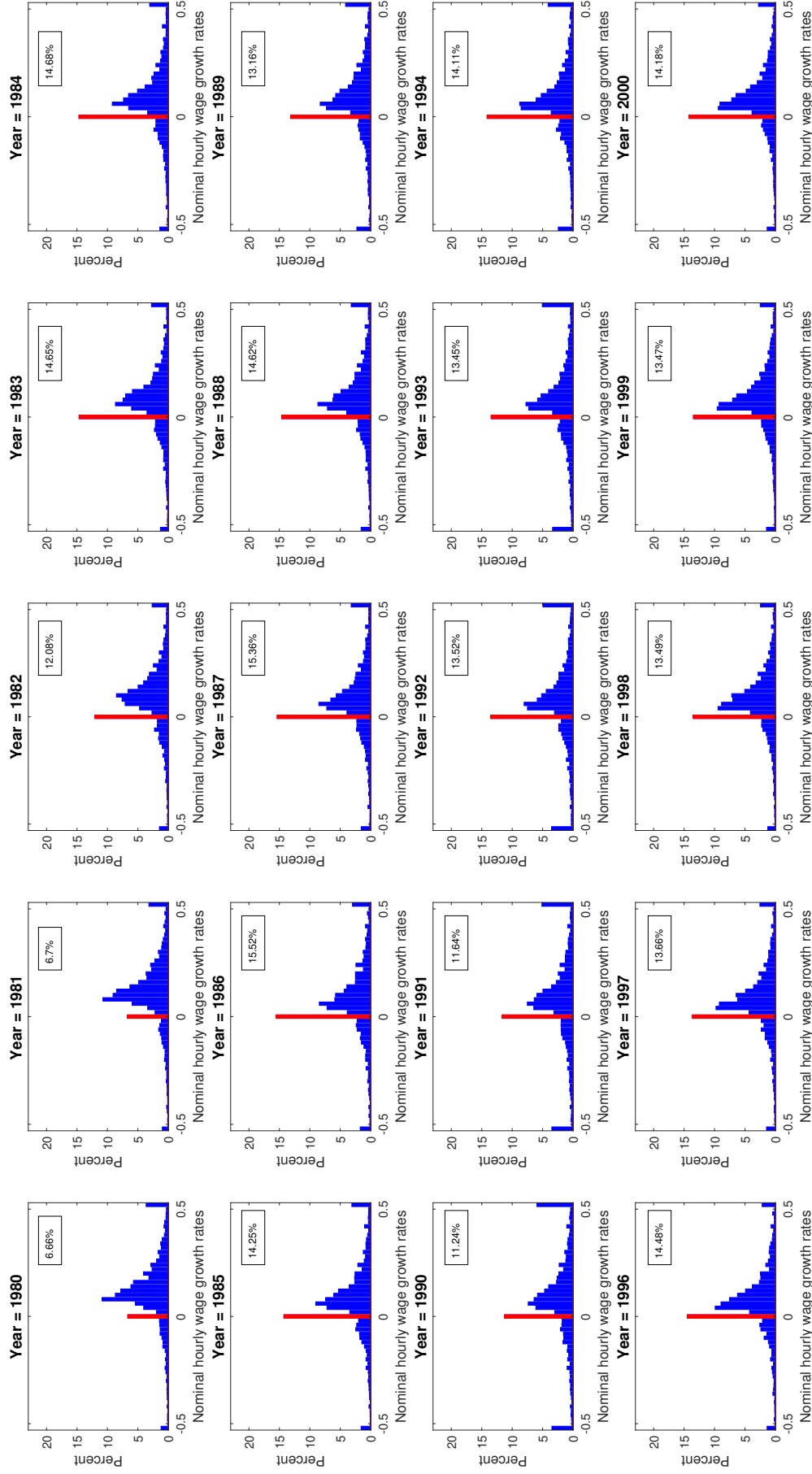


Figure A2: Nominal hourly wage growth rate distributions from 1980 to 2000

Data source: CPS and author's calculation. This figure illustrates year-over-year hourly wage change distributions for hourly paid workers for each year from 1980-2000. The red bin shows the spike at zero, the percentage of workers whose hourly wage growth rate is exactly zero. The width of blue bin is 0.02.

Hourly paid workers, CPS, 2001-2018

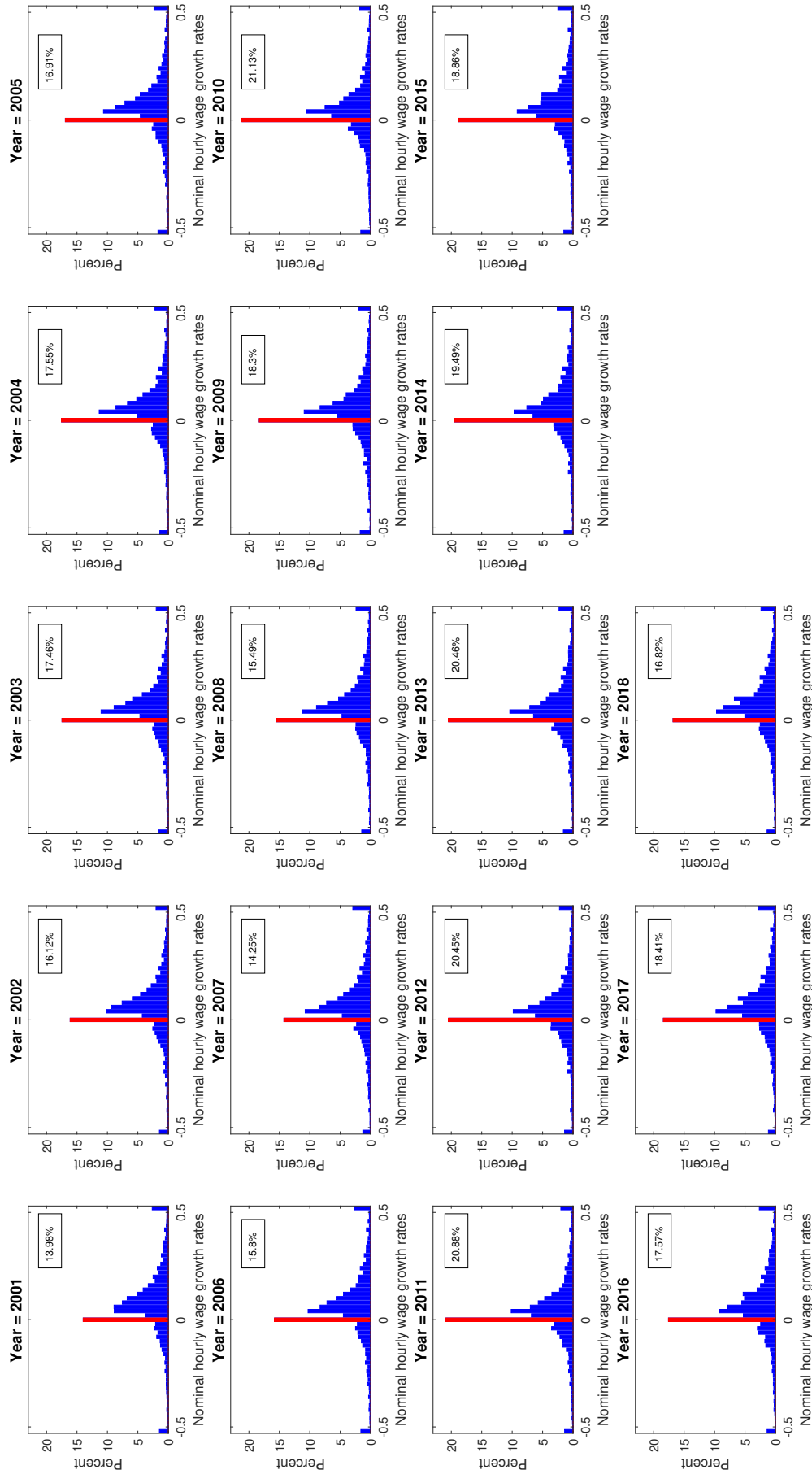


Figure A3: Nominal hourly wage growth rate distributions from 2001 to 2018

Data source: CPS and author's calculation. This figure illustrates year-over-year hourly wage change distributions for hourly paid workers for each year from 2001-2018. The red bin shows the spike at zero, the percentage of workers whose hourly wage growth rate is exactly zero. The width of blue bin is 0.02.

In contrast, clear heterogeneity in nominal hourly wage change distributions can be observed when segmented by hourly wage level and industry. Table A5 presents the averages of the spike at zero, and the share of workers with wage cuts and raises for different income quartiles. Workers in the lower income quartile tend to exhibit a larger spike at zero and a larger share of wage cuts compared to those in the higher income quartiles. Further, Table A6 presents the averages, calculated separately for workers in each 2-digit NAICS industry code, sorted by the average size of the spike at zero. The spike at zero varies widely across industries, ranging from 11 percent to 23 percent. The manufacturing sector employs the highest number of hourly workers in the US and exhibits an average spike at zero of about 14 percent, aligning closely with the national average.

Table A5: Nominal hourly wage change distribution, CPS, by hourly wage quartiles

Hourly wage Quartiles	Spike at zero $\Delta W = 0$	Share of cut $\Delta W < 0$	Share of raises $\Delta W > 0$
25th below	20.88	31.61	47.50
25th to med	15.50	20.70	63.79
med to 75th	13.33	18.09	68.58
75th and above	12.90	16.68	70.42

Data source: CPS and author's calculation. Sample Period: 1979-2018 (except 1995). This table shows the sample average of the spike at zero and the fraction of workers with wage cuts and raises over time by hourly wage quartiles.

A.2.2 Robustness checks for aggregate time series results

The aggregate time series findings are robust to various worker subgroups. For salaried workers, hourly wages can be calculated by dividing usual weekly earnings by the number of usual weekly hours worked. Table A7 displays regression results utilizing these computed hourly wages for salaried workers. It's evident that the spike at zero still exhibits a negative correlation with both inflation and employment levels. Furthermore, the spike at zero shows a stronger association with employment than the share of workers experiencing wage cuts.

Table A8 presents regression results using the regression equation (1), with the exclusion of minimum wage workers. Similarly, Table A9 demonstrates the results based on equation (1), when considering only the working-age population, ranging from 16 to 54 years old. The main findings prove robust even when we exclude minimum wage workers or focus solely on the working-age demographic.

Table A10 demonstrates regression results based on the regression equation (1), cate-

Table A6: The average of the spike at zero, the share of wage cuts and raises by industry, CPS

	% hourly workers	Spike at zero $\Delta W = 0$	Share of cuts $\Delta W < 0$	Share of raises $\Delta W > 0$
Agriculture, Forestry, Fishing and Hunting	1.05	23.82	20.85	55.32
Other Services (except Public Administration)	3.70	22.09	22.04	55.87
Administrative and Support and Waste Management and Remediation Services	1.61	20.57	23.32	56.10
Real Estate and Rental and Leasing	0.96	18.48	20.35	61.17
Arts, Entertainment, and Recreation	1.87	18.31	22.78	58.91
Accommodation and Food Services	7.69	18.15	26.20	55.66
Construction	6.43	17.75	21.07	61.19
Professional, Scientific, and Technical Services	3.27	17.72	17.73	64.55
Wholesale Trade	3.07	16.28	19.52	64.20
Retail Trade	14.53	15.83	20.52	63.65
Educational Services	5.19	14.72	21.77	63.50
Mining, Quarrying, and Oil and Gas Extraction	0.71	14.34	23.82	61.85
Transportation and Warehousing	4.53	13.71	22.76	63.53
Manufacturing	20.70	13.66	20.78	65.56
Health Care and Social Assistance	15.12	13.32	19.59	67.09
Finance and Insurance	2.67	12.75	18.69	68.55
Information	1.42	12.14	20.48	67.38
Utilities	1.68	11.63	20.11	68.25
Public Administration	3.81	11.21	19.97	68.82

Data source: CPS and author's calculation. Sample Period: 1979-2018 (except 1995). This table shows the average of the spike at zero and the fraction of workers with wage cuts and raises over time by 2 digit NAICS industry classification.

gorizing by different levels of education. Likewise, Tables [A11](#), [A12](#), [A13](#), [A14](#) report regression results according to various characteristics, such as age, gender, race, and hourly wage quartiles, respectively. The main results also remain consistent across different worker characteristics, demonstrating that (i) the spike at zero increases when employment falls, after adjusting for inflation, and (ii) the increase in the spike at zero is greater than the rise in the share of wage cuts when employment drops.

Lastly, Table [A15](#) shows regression results for low-paid young workers, who are unlikely to be under a long-term contract. Those hourly workers under the age of 30, earning less than the 25th percentile of hourly wages but more than minimum wages, are classified as low-paid young workers. This subset of workers also displays the countercyclical fluctuations of the spike at zero and the share of workers experiencing wage cuts.

Table A7: The spike at zero, the share of wage cuts, and raises for salaried workers along business cycles

	(1) Spike at zero $\Delta W = 0$	(2) Share of cuts $\Delta W < 0$	(3) Share of raises $\Delta W > 0$	(4) Spike at zero minus Share of cuts	(5) Spike at zero $\Delta W = 0$	(6) Share of cuts $\Delta W < 0$	(7) Share of raises $\Delta W > 0$	(8) Spike at zero minus Share of cuts
1-Epop ratio ($1 - e_t$)	0.432*** (0.134)	-0.0627 (0.438)	-0.369 (0.464)	0.495 (0.354)	0.474*** (0.0538)	0.0521 (0.255)	-0.526* (0.294)	0.421* (0.222)
Inflation rate (π_t)					-0.282*** (0.0233)	-0.779*** (0.0745)	1.061*** (0.0806)	0.496*** (0.0756)
Observations	37	37	37	37	37	37	37	37
Adjusted R^2	0.413	-0.0263	0.0246	0.117	0.656	0.434	0.600	0.292

Data source: CPS and author's calculation. Sample Period: 1979-2018 (except 1994, 1995). The inflation rate is calculated from CPI-U. Hourly rate is calculated from usual weekly earning/usual hours worked per week. Controlling for inflation, the spike at zero exhibits countercyclical fluctuations in employment while the share of workers with wage cuts does not respond to employment. Heteroskedasticity and autocorrelation robust standard errors are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A8: Excluding minimum wage workers, the spike at zero, the share of wage cuts, and raises

	(1) Spike at zero $\Delta W = 0$	(2) Share of cuts $\Delta W < 0$	(3) Share of raises $\Delta W > 0$	(4) Spike at zero minus Share of cuts	(5) Spike at zero $\Delta W = 0$	(6) Share of cuts $\Delta W < 0$	(7) Share of raises $\Delta W > 0$	(8) Spike at zero minus Share of cuts
1-Epop ratio ($1 - e_t$)	0.369 (0.653)	0.197 (0.453)	-0.566 (0.932)	0.172 (0.395)	0.557* (0.287)	0.300 (0.268)	-0.857* (0.487)	0.257 (0.282)
Inflation rate (π_t)					-1.239*** (0.0759)	-0.675*** (0.157)	1.915*** (0.142)	-0.564** (0.224)
Observations	38	38	38	38	38	38	38	38
Adjusted R^2	0.0172	-0.00488	0.0170	-0.0149	0.679	0.332	0.686	0.144

Data source: CPS and author's calculation. Sample Period: 1979-2018 (except 1995). The inflation rate is calculated from CPI-U. The spike at zero, the share of wage cuts and raises are constructed among hourly paid workers whose hourly wages are not subject to minimum wage. Heteroskedasticity and autocorrelation robust standard errors are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A9: The spike at zero, the share of wage cuts, and raises among prime-aged hourly workers along the business cycles

	(1) Spike at zero $\Delta W = 0$	(2) Share of cuts $\Delta W < 0$	(3) Share of raises $\Delta W > 0$	(4) Spike at zero minus Share of cuts	(5) Spike at zero $\Delta W = 0$	(6) Share of cuts $\Delta W < 0$	(7) Share of raises $\Delta W > 0$	(8) Spike at zero minus Share of cuts
1-Epop ratio ($1 - e_t$)	0.0595 (0.586)	0.0353 (0.508)	-0.0948 (0.945)	0.0242 (0.265)	0.457* (0.245)	0.264 (0.285)	-0.721 (0.497)	0.192 (0.144)
Inflation rate (π_t)					-1.248*** (0.0990)	-0.719*** (0.140)	1.966*** (0.144)	-0.529*** (0.192)
Observations	38	38	38	38	38	38	38	38
Adjusted R^2	-0.0261	-0.0269	-0.0261	-0.0274	0.693	0.304	0.675	0.104

Data source: CPS and author's calculation. Sample Period: 1979-2018 (except 1995). The inflation rate is calculated from CPI-U. The spike at zero, the share of wage cuts and raises are constructed among prime-aged hourly paid workers. Heteroskedasticity and autocorrelation robust standard errors are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A10: The spike at zero, the share of wage cuts, and raises by education

	High School or less				College or more			
	(1) Spike at zero $\Delta W = 0$	(2) Share of cuts $\Delta W < 0$	(3) Share of raises $\Delta W > 0$	(4) Spike at zero minus Share of cuts	(5) Spike at zero $\Delta W = 0$	(6) Share of cuts $\Delta W < 0$	(7) Share of raises $\Delta W > 0$	(8) Spike at zero minus Share of cuts
1-Epop	0.551** (0.220)	0.296 (0.262)	-0.847** (0.387)	0.255 (0.198)	0.666*** (0.249)	0.324 (0.257)	-0.989** (0.443)	0.342 (0.223)
Inflation	-1.188*** (0.0737)	-0.717*** (0.162)	1.906*** (0.196)	-0.471*** (0.138)	-1.235*** (0.0659)	-0.628*** (0.134)	1.863*** (0.132)	-0.606*** (0.169)
Observations	38	38	38	38	38	38	38	38
Adjusted R^2	0.699	0.353	0.688	0.112	0.713	0.315	0.696	0.197

Data source: CPS and author's calculation. The inflation rate is calculated from CPI-U. Sample Period: 1979-2018 (except 1995). The first four columns show the regression results for hourly paid workers with high school or less degree, and the last four columns show the regression results for hourly paid workers with college or higher degree. Heteroskedasticity and autocorrelation robust standard errors are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A11: The spike at zero, the share of wage cuts, and raises by age

	16 <= age < 40				40 <= age < 64			
	(1) Spike at zero $\Delta W = 0$	(2) Share of cuts $\Delta W < 0$	(3) Share of raises $\Delta W > 0$	(4) Spike at zero minus Share of cuts	(5) Spike at zero $\Delta W = 0$	(6) Share of cuts $\Delta W < 0$	(7) Share of raises $\Delta W > 0$	(8) Spike at zero minus Share of cuts
1-Epop	0.578*** (0.144)	0.245 (0.209)	-0.823** (0.348)	0.333** (0.152)	0.616*** (0.203)	0.356 (0.334)	-0.971** (0.459)	0.260 (0.266)
Inflation	-1.091*** (0.105)	-0.697*** (0.135)	1.788*** (0.172)	-0.393*** (0.111)	-1.181*** (0.0553)	-0.609*** (0.193)	1.790*** (0.208)	-0.571*** (0.189)
Observations	38	38	38	38	38	38	38	38
Adjusted R^2	0.738	0.391	0.675	0.185	0.716	0.214	0.680	0.0981

Data source: CPS and author's calculation. The inflation rate is calculated from CPI-U. Sample Period: 1979-2018 (except 1995). The first four columns show the regression results for hourly paid workers whose ages are greater than 16 and less than 40, and the last four columns show the regression results for hourly paid workers whose ages are greater than 40 and less than 64. Heteroskedasticity and autocorrelation robust standard errors are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A12: The spike at zero, the share of wage cuts, and raises by gender

	Male				Female			
	(1) Spike at zero $\Delta W = 0$	(2) Share of cuts $\Delta W < 0$	(3) Share of raises $\Delta W > 0$	(4) Spike at zero minus Share of cuts	(5) Spike at zero $\Delta W = 0$	(6) Share of cuts $\Delta W < 0$	(7) Share of raises $\Delta W > 0$	(8) Spike at zero minus Share of cuts
1-Epop	0.515** (0.239)	0.340 (0.266)	-0.855** (0.411)	0.175 (0.288)	0.716*** (0.195)	0.252 (0.269)	-0.968** (0.389)	0.463** (0.187)
Inflation	-1.102*** (0.0652)	-0.506** (0.197)	1.608*** (0.210)	-0.596*** (0.224)	-1.264*** (0.0696)	-0.878*** (0.123)	2.142*** (0.170)	-0.387*** (0.0986)
Observations	38	38	38	38	38	38	38	38
Adjusted R^2	0.674	0.190	0.622	0.138	0.758	0.463	0.735	0.162

Data source: CPS and author's calculation. The inflation rate is calculated from CPI-U. Sample Period: 1979-2018 (except 1995). The first four columns show the regression results for male hourly paid workers, and the last four columns show the regression results for female hourly paid workers. Heteroskedasticity and autocorrelation robust standard errors are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A13: The spike at zero, the share of wage cuts, and raises by race

	White				Non-white			
	(1) Spike at zero $\Delta W = 0$	(2) Share of cuts $\Delta W < 0$	(3) Share of raises $\Delta W > 0$	(4) Spike at zero minus Share of cuts	(5) Spike at zero $\Delta W = 0$	(6) Share of cuts $\Delta W < 0$	(7) Share of raises $\Delta W > 0$	(8) Spike at zero minus Share of cuts
1-Epop	0.632*** (0.223)	0.333 (0.244)	-0.964** (0.398)	0.299* (0.173)	0.554*** (0.190)	0.0753 (0.326)	-0.629* (0.367)	0.479 (0.334)
Inflation	-1.200*** (0.0616)	-0.677*** (0.147)	1.877*** (0.175)	-0.523*** (0.131)	-1.077*** (0.0946)	-0.589** (0.233)	1.666*** (0.222)	-0.489* (0.252)
Observations	38	38	38	38	38	38	38	38
Adjusted R^2	0.739	0.368	0.710	0.182	0.615	0.149	0.623	0.0672

Data source: CPS and author's calculation. The inflation rate is calculated from CPI-U. Sample Period: 1979-2018 (except 1995). The first four columns show the regression results for white hourly paid workers, and the last four columns show the regression results for non-white hourly paid workers. Heteroskedasticity and autocorrelation robust standard errors are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A14: The spike at zero, the share of wage cuts and raises by hourly wage quartiles

	25th below				From 25th to Median			
	(1) Spike at zero $\Delta W = 0$	(2) Share of cuts $\Delta W < 0$	(3) Share of raises $\Delta W > 0$	(4) Spike at zero minus Share of cuts	(5) Spike at zero $\Delta W = 0$	(6) Share of cuts $\Delta W < 0$	(7) Share of raises $\Delta W > 0$	(8) Spike at zero minus Share of cuts
1-Epop	0.972*** (0.291)	0.221 (0.226)	-1.193*** (0.366)	0.751** (0.372)	0.622** (0.314)	0.128 (0.365)	-0.750 (0.539)	0.494* (0.267)
Inflation	-1.246*** (0.343)	-0.935*** (0.318)	2.181*** (0.651)	-0.310*** (0.115)	-1.216*** (0.0896)	-0.685*** (0.153)	1.900*** (0.155)	-0.531*** (0.169)
Observations	38	38	38	38	38	38	38	38
Adjusted R^2	0.494	0.291	0.485	0.127	0.586	0.195	0.540	0.111

	Median to 75th				Above 75th			
	(1) Spike at zero $\Delta W = 0$	(2) Share of cuts $\Delta W < 0$	(3) Share of raises $\Delta W > 0$	(4) Spike at zero minus Share of cuts	(5) Spike at zero $\Delta W = 0$	(6) Share of cuts $\Delta W < 0$	(7) Share of raises $\Delta W > 0$	(8) Spike at zero minus Share of cuts
1-Epop	0.426 (0.331)	0.381 (0.262)	-0.807 (0.553)	0.0453 (0.271)	0.552* (0.285)	0.439* (0.236)	-0.992** (0.479)	0.113 (0.247)
Inflation	-1.114*** (0.101)	-0.403*** (0.118)	1.517*** (0.125)	-0.711*** (0.192)	-1.150*** (0.100)	-0.703*** (0.155)	1.852*** (0.128)	-0.447** (0.220)
Observations	38	38	38	38	38	38	38	38
Adjusted R^2	0.537	0.193	0.533	0.205	0.661	0.428	0.718	0.0829

Data source: CPS and author's calculation. Sample Period: 1979-2018 (except 1995). This table shows the cyclicity of the spike at zero, the share of wage cuts and raises by hourly wage quartiles. The inflation rate is calculated from CPI-U. Heteroskedasticity and autocorrelation robust standard errors are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Heteroskedasticity-robust standard errors are in parentheses.

Table A15: The spike at zero, the share of wage cuts, and raises for low-paid young workers along the business cycles

	(1) Spike at zero $\Delta W = 0$	(2) Share of cuts $\Delta W < 0$	(3) Share of raises $\Delta W > 0$	(4) Spike at zero minus Share of cuts	(5) Spike at zero $\Delta W = 0$	(6) Share of cuts $\Delta W < 0$	(7) Share of raises $\Delta W > 0$	(8) Spike at zero minus Share of cuts
1-Epop ratio ($1 - e_t$)	0.702 (0.570)	0.806*** (0.260)	-1.508*** (0.503)	-0.104 (0.694)	0.899*** (0.167)	0.884** (0.410)	-1.783*** (0.347)	0.0146 (0.502)
Inflation rate (π_t)					-1.298*** (0.0914)	-0.513 (0.484)	1.811*** (0.537)	-0.784* (0.421)
Observations	38	38	38	38	38	38	38	38
Adjusted R^2	0.105	0.0986	0.161	-0.0259	0.696	0.142	0.509	0.0918

Data source: CPS and author's calculation. Sample Period: 1979-2017 (except 1995). The inflation rate is calculated from CPI-U. The spike at zero, the share of workers with raises and cuts come from the annual nominal hourly wage growth distribution of low-paid young workers, who are younger than the age of 30 and earn less than equal to the 25 percentile of hourly wages for each year and greater than the minimum wages. Heteroskedasticity and autocorrelation robust standard errors are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

A.2.3 Robustness checks for state-level results

Table A16: Changes in nominal wage distribution from 1979 to 1982 across states

	(1)	(2)	(3)	(4)	(5)
	Changes in Spike at zero $\Delta W = 0$	Changes in Share of cuts $\Delta W < 0$	Changes in Share of raises $\Delta W > 0$	Changes in Spike at zero minus Share of cuts	$\ln \frac{W_{s1982}}{W_{s1979}}$
Percent change in employment rate	-0.271 (0.277)	0.296 (0.372)	-0.0248 (0.445)	-0.567 (0.481)	0.531** (0.249)
Observations	47	47	47	47	47
Adjusted R^2	-0.00507	-0.00607	-0.0222	0.00774	0.0427

Data source: CPS and author's calculation. Sample Period: 1979 - 1982. This table shows changes in nominal wage change distributions along with employment for each state from 1979 to 1982. Heteroskedasticity-robust standard errors are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A17: The state-level regression results

State	Dependent Variable		State	Dependent Variable	
	Spike at zero	Spike at Zero minus Share of Cuts		Spike at zero	Spike at Zero minus Share of Cuts
Alabama	0.36*	0.07	Montana	-0.03	-0.38
	(0.20)	(0.29)		(0.17)	(0.39)
Alaska	0.48*	0.51	Nebraska	0.47**	0.77***
	(0.28)	(0.38)		(0.20)	(0.25)
Arizona	0.71**	-0.14	Nevada	0.27	-0.17
	(0.27)	(0.25)		(0.16)	(0.24)
Arkansas	0.41*	0.32	New Hampshire	1.29***	0.94*
	(0.24)	(0.33)		(0.27)	(0.50)
California	0.99***	0.30	New Jersey	0.51	0.38
	(0.21)	(0.21)		(0.51)	(0.55)
Colorado	0.46*	0.31	New Mexico	0.53**	-0.19
	(0.23)	(0.29)		(0.24)	(0.27)
Connecticut	0.96**	0.98*	New York	-0.24	-0.76
	(0.43)	(0.58)		(0.41)	(0.49)
Delaware	0.52**	0.67**	North Carolina	0.84***	0.53**
	(0.25)	(0.30)		(0.12)	(0.24)
Florida	0.43	0.44	North Dakota	0.11	0.42***
	(0.50)	(0.34)		(0.18)	(0.15)
Georgia	0.71***	0.52**	Ohio	0.12	-0.33*
	(0.18)	(0.24)		(0.21)	(0.19)
Hawaii	0.38	-0.16	Oklahoma	0.71	-0.62
	(0.29)	(0.40)		(0.53)	(0.63)
Idaho	0.45**	0.15	Oregon	0.76***	0.44
	(0.19)	(0.19)		(0.19)	(0.30)
Illinois	0.05	-0.16	Pennsylvania	-0.14	-0.29*
	(0.31)	(0.34)		(0.20)	(0.15)
Indiana	0.40	0.27	Rhode Island	0.33	0.14
	(0.24)	(0.25)		(0.39)	(0.27)
Iowa	0.51***	0.23	South Carolina	0.52***	0.42
	(0.17)	(0.16)		(0.17)	(0.27)
Kansas	0.84***	0.57**	South Dakota	0.19	0.19
	(0.30)	(0.27)		(0.20)	(0.25)
Kentucky	0.40	-0.13	Tennessee	0.51*	0.42
	(0.28)	(0.34)		(0.28)	(0.29)
Louisiana	1.61***	0.64	Texas	1.12***	0.74**
	(0.39)	(0.62)		(0.20)	(0.29)
Maine	0.24	0.30	Utah	0.55**	0.39*
	(0.38)	(0.31)		(0.24)	(0.20)
Maryland	1.00*	1.09*	Vermont	-0.01	-0.11
	(0.56)	(0.59)		(0.24)	(0.30)
Massachusetts	0.82**	-0.03	Virginia	0.82**	1.39***
	(0.37)	(0.57)		(0.36)	(0.54)
Michigan	0.53***	0.23*	Washington	0.30	0.32
	(0.18)	(0.13)		(0.25)	(0.25)
Minnesota	0.29	0.16	West Virginia	0.29*	0.28
	(0.20)	(0.22)		(0.15)	(0.17)
Mississippi	0.20	0.21	Wisconsin	0.63**	0.45**
	(0.27)	(0.40)		(0.29)	(0.21)
Missouri	0.23	0.03	Wyoming	0.76 ***	-0.43
	(0.23)	(0.20)		(0.28)	(0.37)

Notes: This table reports regression estimates for the following state-specific regressions:

$$\begin{aligned}
 [\text{Spike at zero}]_{it} &= \alpha_1 + \beta_1(1 - e_{it}) + \epsilon_{it} \\
 [\text{Spike at zero- Share of cuts}]_{it} &= \alpha_2 + \beta_2(1 - e_{it}) + \epsilon_{it} \quad ,
 \end{aligned}$$

where e_{it} represents state-level employment to population ratio. Standard errors are in parentheses. * p<0.10, ** p<0.05, ***p<0.01.

A.3 Appendix: SIPP

This section present empirical results using the SIPP data. I begin with constructing the three key statistics - the spike at zero and the share of workers with wage cuts and raises. Table A18 presents the unweighted count of observations of hourly workers whose hourly wage growth rate is available for each year and the time series of the three statistics. Table A19 divides hourly workers into two categories - job stayers and job switchers - and provides the unweighted count of observations, the three statistics for each group.

Table A18: The spike at zero, the share of wage cuts, and raises in the SIPP

Year	Obs ΔW	Spike at zero (%) $\Delta W = 0$	Share of cuts (%) $\Delta W < 0$	Share of raises (%) $\Delta W > 0$
1985	9,827	16.75	18.76	64.50
1986	13,490	17.26	19.36	63.38
1987	11,171	17.92	20.11	61.97
1988	10,508	14.95	18.12	66.93
1989	10,930	14.63	17.92	67.44
1991	11,820	14.30	18.74	66.96
1992	17,241	17.31	19.32	63.37
1993	16,318	18.58	20.29	61.14
1994	19,430	18.28	20.66	61.07
1995	9,347	18.31	18.58	63.12
1997	16,951	14.02	18.68	67.30
1998	15,877	14.31	16.33	69.37
1999	14,939	16.98	16.91	66.11
2000	5,408	17.52	15.29	67.20
2002	13,727	16.12	21.85	62.04
2003	12,287	19.27	19.51	61.21
2005	20,055	30.13	17.31	52.57
2006	17,621	30.05	14.19	55.76
2007	7,922	31.48	13.64	54.88
2009	13,909	39.85	16.85	43.29
2010	16,080	42.22	16.00	41.77
2011	14,228	45.59	13.24	41.17
2012	13,242	43.84	13.72	42.44
2013	11,943	46.46	12.61	40.93
Mean	13,511	24.00	17.42	58.58

Data source: SIPP and author's calculation. Sample period: 1984 - 2013 except 1990, 1996, 2001, and 2008. This table shows the unweighted number of observation and the size of peak, the share of workers with wage cuts and raises for hourly paid workers.

Table A19: The spike at zero, the share of wage cuts, and raises in the SIPP by job stayers and job switchers

Year	Job stayers				Job switchers			
	Obs	Spike	Share of	Share of	Obs	Spike at	Share of	Share of
	ΔW	zero (%) $\Delta W = 0$	cuts(%) $\Delta W < 0$	raises(%) $\Delta W > 0$	ΔW	zero (%) $\Delta W = 0$	cuts(%) $\Delta W < 0$	raises(%) $\Delta W > 0$
1985	7,724	16.95	16.08	66.97	2,103	15.99	28.52	55.49
1986	9,735	18.58	16.14	65.28	3,755	13.50	28.50	58.00
1987	8,489	19.46	16.80	63.74	2,682	12.88	30.96	56.16
1988	7,593	16.70	14.00	69.30	2,915	10.35	28.92	60.73
1989	7,949	16.45	14.09	69.46	2,981	9.66	28.44	61.90
1991	8,699	16.41	13.70	69.89	3,121	8.43	32.78	58.79
1992	13,226	19.30	15.02	65.67	4,015	10.70	33.52	55.77
1993	12,514	20.97	16.34	62.69	3,804	10.66	33.36	55.98
1994	14,422	20.64	16.54	62.82	5,008	11.54	32.39	56.07
1995	6,935	20.56	14.92	64.52	2,412	11.86	29.03	59.11
1997	11,184	16.20	14.84	68.96	5,767	9.86	26.04	64.11
1998	10,290	17.05	12.05	70.91	5,587	9.30	24.16	66.55
1999	9,851	19.71	12.38	67.91	5,088	11.73	25.61	62.66
2000	3,938	20.00	11.54	68.45	1,470	10.93	25.20	63.87
2002	8,926	18.92	16.34	64.74	4,801	10.91	32.06	57.03
2003	8,491	22.17	14.25	63.57	3,796	12.81	31.25	55.94
2005	13,282	38.87	10.14	50.99	6,773	13.29	31.10	55.61
2006	11,937	38.60	7.42	53.98	5,684	12.75	27.90	59.35
2007	5,339	40.88	6.81	52.31	2,583	12.04	27.78	60.18
2009	10,194	49.10	10.21	40.69	3,715	15.44	34.41	50.16
2010	11,292	53.83	8.44	37.73	4,788	15.92	33.15	50.93
2011	10,076	57.39	6.46	36.15	4,152	18.01	29.08	52.92
2012	9,333	56.21	6.21	37.58	3,909	15.84	30.73	53.43
2013	8,695	58.39	5.07	36.54	3,248	16.18	31.75	52.08
Mean	9,588	28.89	12.32	58.79	3,923	12.52	29.86	57.62

Data source: SIPP and author's calculation. Sample period: 1984 - 2013 except 1990, 1996, 2001, and 2008. This table shows the number of observation and the spike at zero, the share of workers with wage cuts and raises for hourly paid job stayers and job switchers.

Hourly paid workers, SIPP, 1985-2013 Job switchers

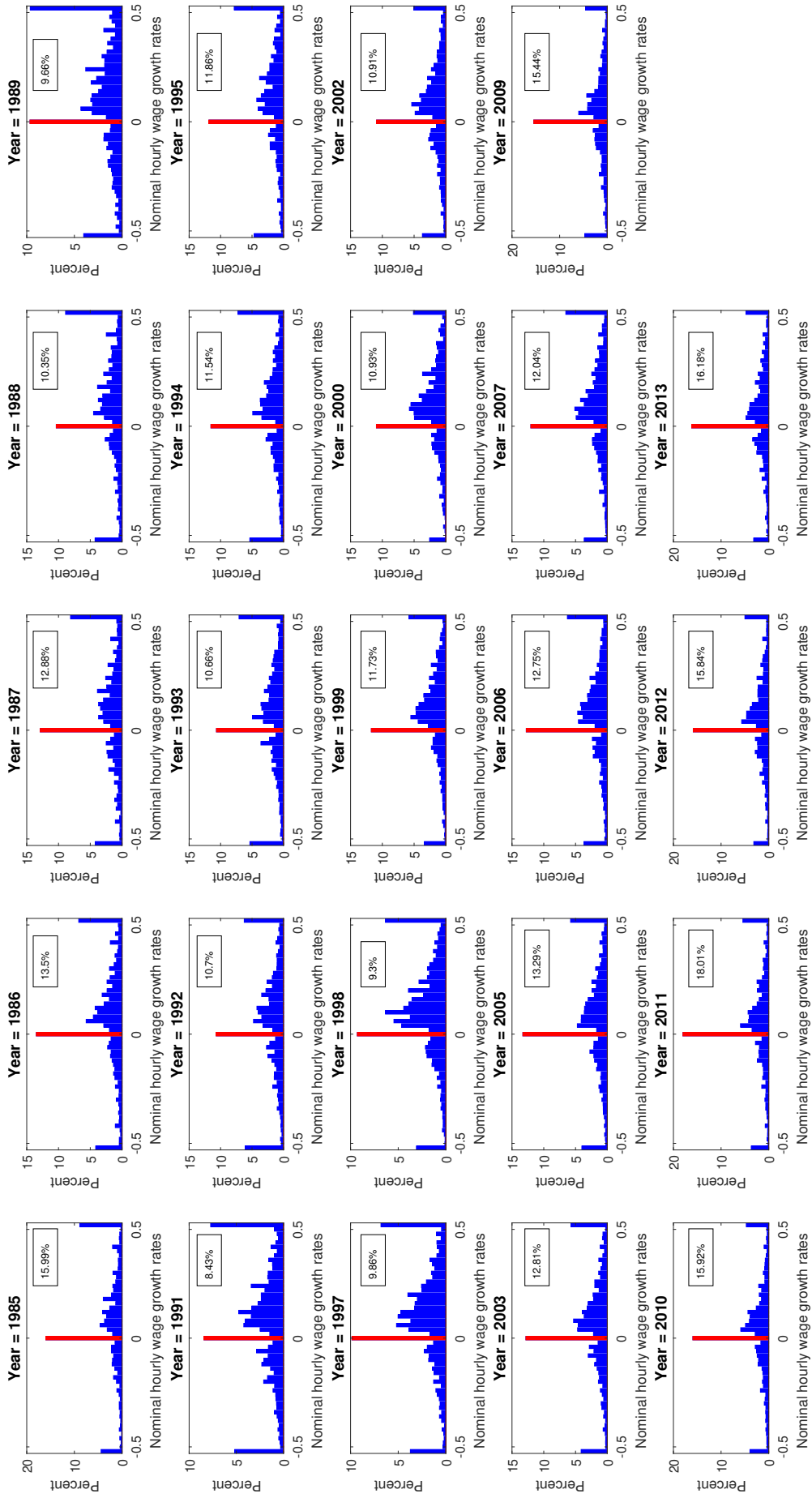


Figure A6: Nominal hourly wage growth rate distributions for 1985-2013 for job switchers

Data source: SIPP and author's calculation. This figure illustrates year-over-year hourly wage change distributions for hourly paid job switchers for each year from 1985-2013 with some gaps. The red bin shows the spike at zero, the percentage of workers whose hourly wage growth rate is exactly zero. Other than the red bin, the width of blue bin is 0.02.

Hourly paid workers, SIPP, 1985-2013

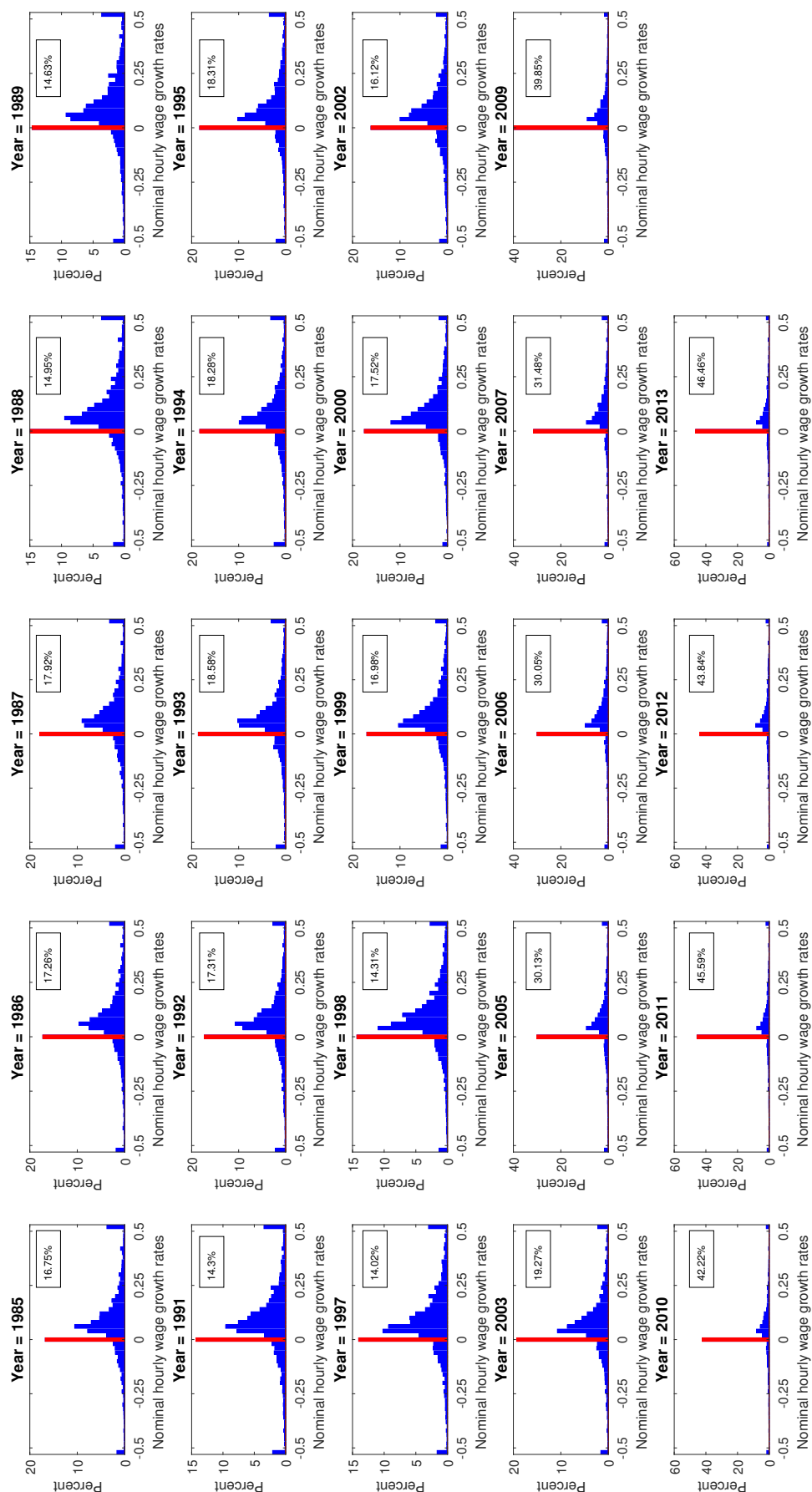


Figure A4: Nominal hourly wage growth rate distributions for 1985-2013

Data source: SIPP and author's calculation. This figure illustrates year-over-year hourly wage change distributions for hourly workers, including both job stayers and job switchers for each year from 1985-2013 with some gaps. The red bin shows the spike at zero, the percentage of workers whose hourly wage growth rate is exactly zero. Other than the red bin, the width of the bin is 0.02.

Hourly paid workers, SIPP, 1985-2013 Job stayers

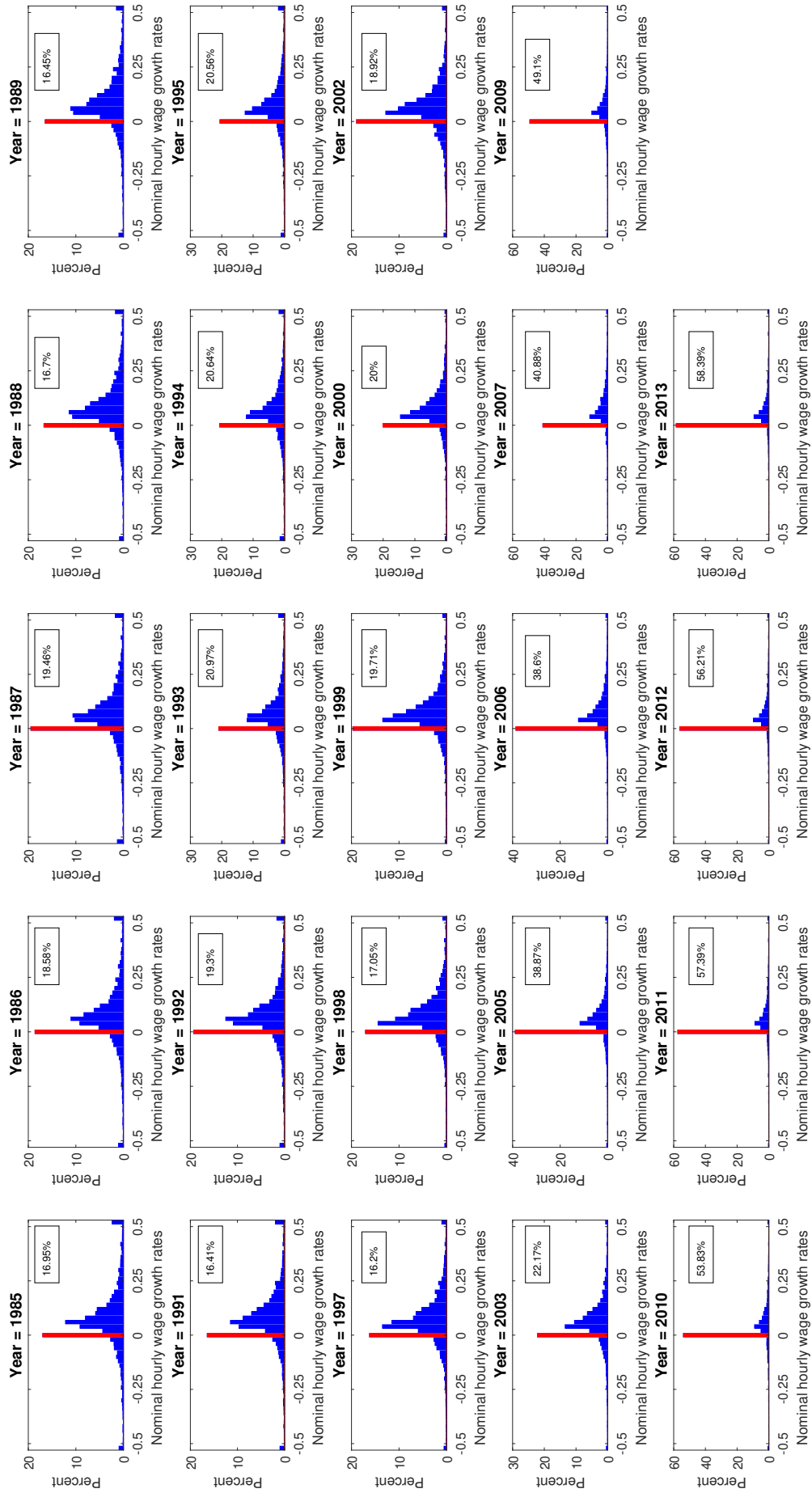


Figure A5: Nominal hourly wage growth rate distributions for 1985-2013 for job stayers

Data source: SIPP and author's calculation. This figure illustrates year-over-year hourly wage change distributions for hourly paid job stayers for each year from 1985-2013 with some gaps. The red bin shows the spike at zero, the percentage of workers whose hourly wage growth rate is exactly zero. Other than the red bin, the width of blue bin is 0.02.

A.3.1 Asymmetric nominal wage change distribution: Job stayers vs. job switchers

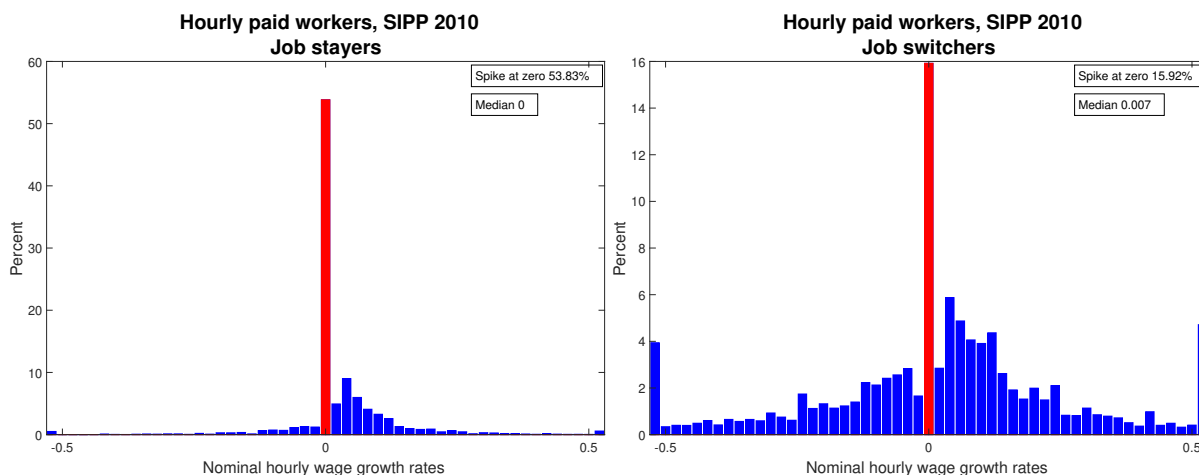


Figure A7: Nominal hourly wage distribution in 2010: job stayers vs. job switchers

Data source: SIPP and author's calculation. The figure shows nominal hourly wage change distribution for job stayers (left) and that for job switchers (right). The red bin shows the spike at zero, which represents the percentage of workers whose hourly wage growth rate is precisely zero from 2009 to 2010. The width of blue bin is 0.02. The bin to the very right includes all the workers whose log nominal hourly wage differences are greater than 0.5, and the bin to the very left includes all the workers whose hourly wage growth rates are less than -0.5.

Nominal wage change distributions constructed from SIPP are also exhibit an asymmetry with a large spike at zero, similar to the findings from the CPS. Figure A4 shows year-over-year hourly wage change distributions for hourly workers including both job stayers and job switchers for each year from 1985-2013 with some gaps. The red bar presents the spike at zero, the share of workers with no wage change and the size of the blue bin is 0.02. Table A20 is also similar to Table A4, reporting sample averages for the share of workers with zero wage changes, wage cuts, and raises. Again, these estimates do not show notable heterogeneity by worker characteristics such as gender and education - common to both the CPS and the SIPP.

.In addition, the SIPP data allows comparing nominal wage change distributions between job stayers and job switchers. I find that the empirical patterns suggestive of DNWR - asymmetry and the spike at zero - are more pronounced for job stayers, but also hold for job switchers. Figure A7 displays nominal hourly wage change distributions in 2010 for job stayers (left) and job switchers (right). Both distributions display large spikes at zero, although the spike for job stayers is much larger than that for job

Table A20: Descriptive statistics by worker characteristics, SIPP

	% of all workers	Spike at zero $\Delta W = 0$	Share of cuts $\Delta W < 0$	Share of raises $\Delta W > 0$
Hourly Paid workers		24.00	17.42	58.58
Exc. Minimum wage workers	89.25	23.99	16.68	59.33
Job stayers	71.08	28.89	12.32	58.79
Job switchers	28.92	12.52	29.86	57.62

Data source: SIPP and author's calculation. Sample Period: 1984-2013 (except 1990, 1996, 2001, 2004, 2008). This table shows the sample average of the spike at zero and the fraction of workers with wage cuts and raises over time by worker characteristics.

switchers.⁴ These characteristics are common to all nominal wage change distributions for each year for the entire sample period, 1985-2013. Figure A5 shows the year-over-year hourly wage change distribution for hourly job stayers and Figure A6 shows that for job switchers for the entire sample period from the SIPP.

Similarly, Table A20 shows that for job stayers, the average size of the spike is larger, whereas the average share of workers with wage cuts is smaller. In fact, the spike at zero for job stayers is always higher than that for job switchers and the share of workers with wage cuts for job stayers is always lower than that for job switchers (See Table A19). These comparisons between job stayers and switchers appear overall consistent with the findings by [Bils \(1985\)](#) and [Shin \(1994\)](#), who argue that wages are more flexible for job switchers than job stayers. However, my findings suggest job switchers' wages may still be downwardly rigid, albeit to a lesser extent. In this regard, [Hazell and Taska \(2024\)](#) also find that posted wages are sticky, leading no wage changes for job switchers.

Because about 71 percent of hourly workers are job stayers in the SIPP, and because nominal hourly wage change distributions for job switchers still exhibit asymmetry and the spike at zero - although to a lesser extent - the distributions using all workers such as Figure A4 exhibit strong asymmetry and a large spike at zero. This is also comparable to Figure A2 and Figure A3, nominal hourly wage change distributions constructed from the CPS, which also includes both job stayers and job switchers, with the former being a large share.

⁴Table A21 in Appendix A.3 shows the average of the spike at zero and the share of wage cuts and raises according to the reasons why hourly workers switched their employer in a given year. Contingent workers or temporarily employed workers, workers on layoff, and injured or ill workers show the high average spike at zero among job switchers.

A.3.2 The wage change distribution for job switchers by reasons for job switching

This section reports the average spike at zero, the share of wage cuts, and increases for job switchers by reasons for it. The SIPP asks the reasons why respondents have stopped working for the previous employer. About 50 percent of job switchers do not respond to this question. Among the other 50 percent, workers on layoff or injured or temporary workers record a higher spike at zero.

Fired/discharged workers present a similar level of the spike at zero compared to workers who quit the job to take another job. However, workers who quit the job to take another job tend to have a higher share of raises and a smaller share of cuts. Fired or discharged workers tend to show a higher share of wage cuts.

Table A21: The spike at zero, the share of wage cuts, and raises for job-switchers by reasons of switching, SIPP

	% of job switchers	Spike at zero $\Delta W = 0$	Fraction of $\Delta W < 0$	Fraction of $\Delta W > 0$
On layoff	11.53	14.06	37.05	48.89
Fired/Discharged	2.35	9.96	43.98	46.07
Quit to take another job	8.27	9.33	22.89	67.78
Contingent worker/temporary employed	4.22	14.38	29.97	55.65
Illness/Injury	1.26	14.26	38.69	47.05
Others	19.54	12.17	32.79	55.04
Missing	52.82	12.23	27.79	59.98

Data source: SIPP and author's calculation. Sample Period: 1984-2013 (except 1990, 1996, 2001, 2004, 2008). This table shows the sample average of the spike at zero and the share of workers with wage cuts and raises over time by reasons of job switching. The category "others" include attending schools, childcare problems, family/personal obligations, unsatisfactory work arrangements, retirement and so on.

A.3.3 The cyclicalness of wage change distributions

A.4 Appendix: Model

A.4.1 Measurement error

In order to compare quantitative results with CPS data, this paper incorporates measurement errors into the simulated data. The distribution of these errors is based on the work of [Bound and Krueger \(1991\)](#), where they compared an individual's reported log earnings from CPS with their true log earnings to compute the measurement error from Social Security payroll tax records. The distributions of measurement errors are illustrated in a

Table A22: The spike at zero, the fraction of wage cuts and raises - job stayers vs. job switchers, SIPP

	All hourly paid workers			Job stayers			Job switchers		
	(1) Spike at zero	(2) Share of cuts	(3) Share of raises	(4) Spike at zero	(5) Share of cuts	(6) Share of raises	(7) Spike at zero	(8) Share of cuts	(9) Share of raises
1-Epop ratio	1.209*** (0.277)	0.148 (0.194)	-1.357*** (0.253)	1.317** (0.523)	0.200 (0.210)	-1.517*** (0.354)	0.464*** (0.124)	0.735*** (0.241)	-1.199*** (0.154)
Inflation rate	-1.678*** (0.545)	-0.369** (0.147)	2.047*** (0.398)	-1.979*** (0.652)	-0.255 (0.198)	2.234*** (0.417)	-0.512** (0.205)	-0.568 (0.398)	1.080*** (0.256)
Panel Fixed Effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	24	24	24	24	24	24	24	24	24
Adjusted R^2	0.934	0.562	0.934	0.939	0.793	0.941	0.613	0.325	0.718

Source: SIPP and author's calculation. Sample Period: 1984-2013 (except 1990, 1996, 2001, 2004, 2008). The first three columns include all hourly workers, columns 4-6 include only job stayers, and last 3 columns include only job switchers. Heteroskedasticity and autocorrelation robust standard errors are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure A8, which replicates Figures 2 and 3 from [Bound and Krueger \(1991\)](#).

The mean of the replicated measurement errors is 0.0043 for men and -0.0167 for women, aligning closely with what [Bound and Krueger \(1991\)](#) report: 0.004 and -0.017, respectively. The variance of these replicated measurement errors is 0.0616 for men and 0.0335 for women, whereas [Bound and Krueger \(1991\)](#) report variances of 0.114 and 0.051, respectively. The discrepancy can be attributed to the fact that [Bound and Krueger \(1991\)](#) only report the trimmed distribution of measurement errors, which naturally limits the variance.

A.4.2 Stationary wage change distribution

A.4.3 Value function

Note that the value function can be written a function of the relative wage rather than both individual and aggregate wage. This representation allows us to reduce one dimension of the problem, following [Nakamura and Steinsson \(2008\)](#).

Under the Calvo wage rigidity, wage setters can optimize their wage with probability $(1 - \mu^{\text{Calvo}})$ regardless of the sign of wage change. To introduce randomness, one more state variable, x_t , a binary variable, is added. Once x_t equals 1 with the probability of $(1 - \mu^{\text{Calvo}})$, wage setters can reoptimize their wage. The recursive problem under the Calvo rigidity can be written as follows:

Table A23: The spike at zero, the fraction of wage cuts and raises - job-stayers vs. job-switchers across states, SIPP

	All hourly paid workers			Job stayers			Job switchers		
	(1) Spike at zero	(2) Share of cuts	(3) Share of raises	(4) Spike at zero	(5) Share of cuts	(6) Share of raises	(7) Spike at zero	(8) Share of cuts	(9) Share of raises
1-Epop ratio ($1 - e_{it}$)	0.428*** (0.0953)	0.0855 (0.0877)	-0.514*** (0.107)	0.513*** (0.118)	0.104 (0.0883)	-0.617*** (0.117)	0.318*** (0.103)	0.139 (0.192)	-0.457** (0.208)
State-level inflation (π_{it})	-1.005 (0.699)	-0.136 (0.475)	1.141 (1.023)	-1.234 (0.765)	-0.0128 (0.212)	1.247 (0.843)	-0.588 (0.501)	-0.0251 (1.745)	0.613 (1.960)
State Fixed Effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Time Fixed Effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Control	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	845	845	845	845	845	845	845	845	845
Adjusted R^2	0.84	0.34	0.79	0.87	0.50	0.82	0.17	0.06	0.15

Data source: SIPP and author's calculation. 43 states included: 36 states for 21 years, 7 states for 20 years. Small states dropped due to small sample sizes. State-level cluster robust standard errors are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

$$\begin{aligned}
& V(q_t(i), L_t, \frac{W_{t-1}(i)}{W_t}, x_t) \\
&= \max_{W_t(i)} \left[H(q_t(i), L_t, \frac{W_t(i)}{W_t}) + \beta \mathbb{E}(V(q_{t+1}(i), L_{t+1}, \frac{W_t(i)}{W_{t+1}}, x_{t+1})) \right] \mathbb{I}(x_t = 1) \\
&+ \max_{W_t(i)} \left[H(q_t(i), L_t, \frac{W_{t-1}(i)}{W_t}) - C \times \mathbb{I}(W_t(i) \neq W_{t-1}(i)) \right. \\
&\quad \left. + \beta \mathbb{E}(V(q_{t+1}(i), L_{t+1}, \frac{W_{t-1}(i)}{W_{t+1}}, x_{t+1})) \right] \mathbb{I}(x_t = 0)
\end{aligned}$$

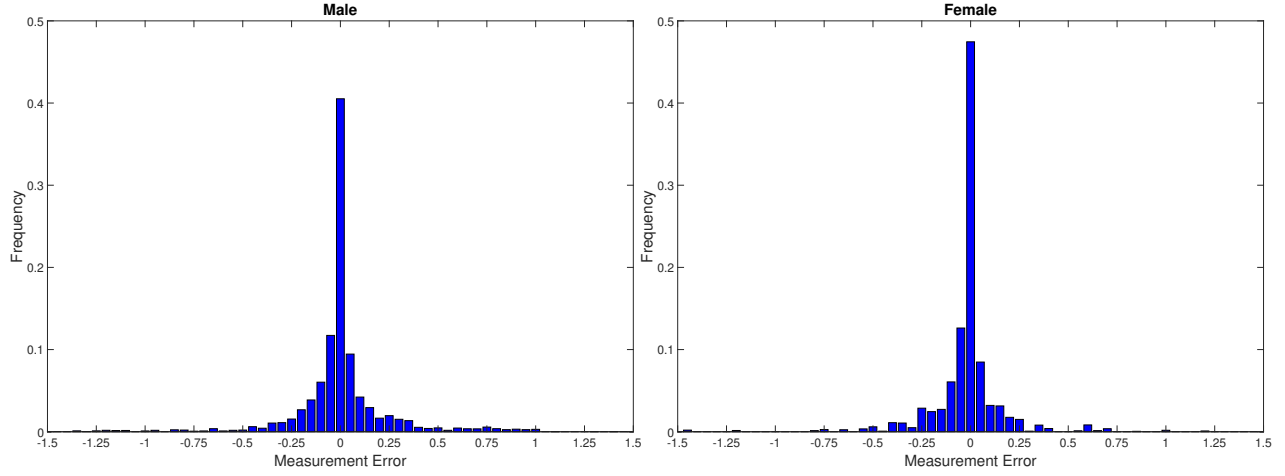
where $C > \infty$ and

$$H(q_t(i), L_t, \frac{w_t(i)}{W_t}) = q_t(i)^{\theta-1} \left(\frac{w_t(i)}{W_t} \right)^{1-\theta} L_t^{(1-\gamma)} - \omega \frac{[q_t(i)^{\theta-1} \left(\frac{w_t(i)}{W_t} \right)^{-\theta} L_t]^{1+\psi}}{1+\psi},$$

which can be derived from substituting labor demand into the households' objective function (Equation (5)). If x_t is one, wage setters adjust nominal wages freely, whereas wage setters must pay infinite cost of wage adjustment if x_t equals zero. These features make the Calvo mode tractable, which is introduced by Nakamura and Steinsson (2008) and called the CalvoPlus model.

For the Symmetric menu cost model, wage setters have to pay an additional fixed cost, K , to adjust their wages with the probability of μ^{Menu} , when x_t equals to zero. With the

Figure A8: Distribution of Measurement error [Bound and Krueger \(1991\)](#)



Notes: The figure on the left replicates Figure 2 from [Bound and Krueger \(1991\)](#), and the figure on the right replicates Figure 3 from [Bound and Krueger \(1991\)](#).

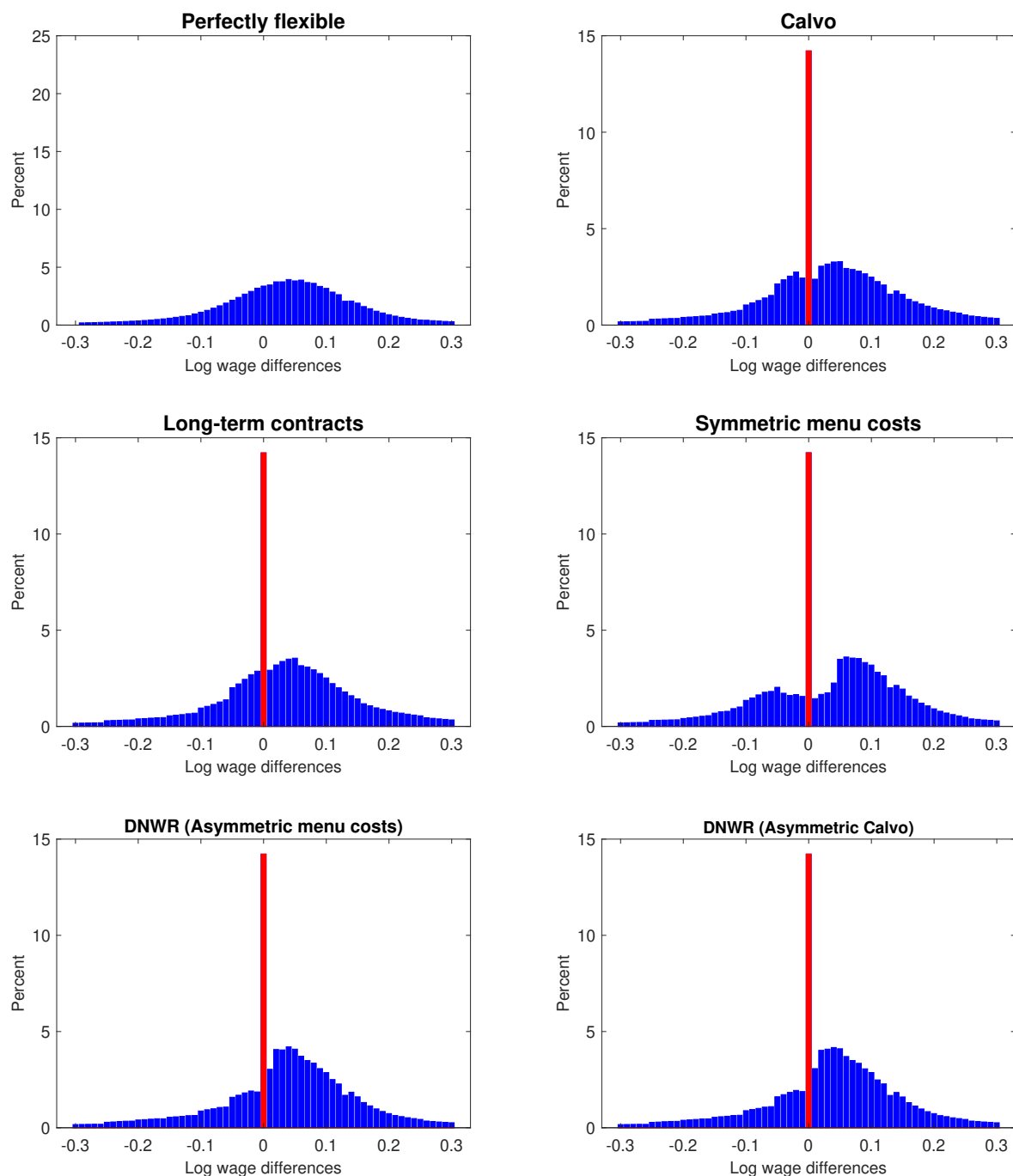
other probability of $(1 - \mu^{\text{Menu}})$, wage setters can adjust wages without any cost. The recursive problem with menu costs can be written as follows:

$$\begin{aligned}
 & V(q_t(i), L_t, \frac{W_{t-1}(i)}{W_t}, x_t) \\
 &= \max_{W_t(i)} \left[H(q_t(i), L_t, \frac{W_t(i)}{W_t}) + \beta \mathbb{E}(V(q_{t+1}(i), L_{t+1}, \frac{W_t(i)}{W_{t+1}}, x_{t+1})) \right] \mathbb{I}(x_t = 1) \\
 &+ \max_{W_t(i)} \left[H(q_t(i), L_t, \frac{W_t(i)}{W_t}) - K \mathbb{I}(W_t(i) \neq W_{t-1}(i)) \right. \\
 &\quad \left. + \beta \mathbb{E}(V(q_{t+1}(i), L_{t+1}, \frac{W_t(i)}{W_{t+1}}, x_{t+1})) \right] \mathbb{I}(x_t = 0).
 \end{aligned}$$

The recursive problem of downward nominal wage rigidity with the Asymmetric menu costs model can be written as follows. For the downward adjustment of nominal wage, wage setters have to pay the fixed cost K^d with the probability of μ^{Menu} .

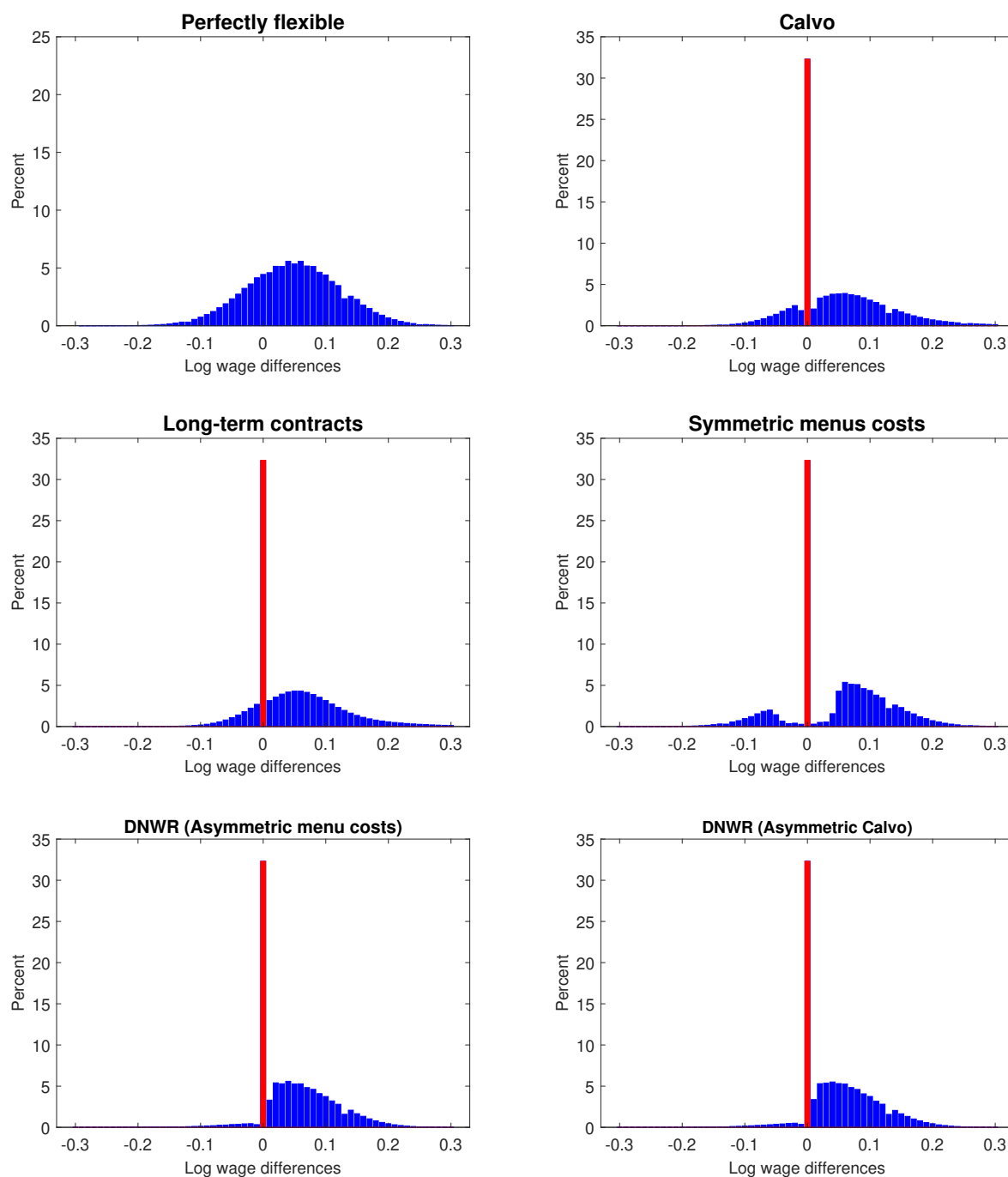
$$\begin{aligned}
 & V(q_t(i), L_t, \frac{W_{t-1}(i)}{W_t}, x_t) \\
 &= \max_{W_t(i)} \left[H(q_t(i), L_t, \frac{W_t(i)}{W_t}) + \beta \mathbb{E}(V(q_{t+1}(i), L_{t+1}, \frac{W_t(i)}{W_{t+1}}, x_{t+1})) \right] \mathbb{I}(x_t = 1) \\
 &+ \max_{W_t(i)} \left[H(q_t(i), L_t, \frac{W_t(i)}{W_t}) - K^d \mathbb{I}(W_t(i) < W_{t-1}(i)) \right. \\
 &\quad \left. + \beta \mathbb{E}(V(q_{t+1}(i), L_{t+1}, \frac{W_t(i)}{W_{t+1}}, x_{t+1})) \right] \mathbb{I}(x_t = 0).
 \end{aligned}$$

Figure A9: Stationary wage change distribution from six wage-setting schemes



Notes: This figure illustrates the stationary wage change distribution with measurement errors, as suggested by six alternative wage-setting models: Perfectly Flexible (top left), Calvo (top right), Long-Term contract (second-row left), Symmetric menu cost (second-row right), Asymmetric menu Cost (bottom left), and Asymmetric Calvo (bottom right). The red bar indicates the proportion of workers with no wage change, while the blue bins, each with a width of 0.01, depict the range of wage changes.

Figure A10: Stationary wage change distribution from six wage-setting schemes without measurement error



Notes: This figure illustrates the stationary wage change distribution without measurement errors, as suggested by six alternative wage-setting models: Perfectly Flexible (top left), Calvo (top right), Long-term Contract (second-row left), Symmetric menu cost (second-row right), Asymmetric menu cost (bottom left), and Asymmetric Calvo (bottom right). The red bar indicates the proportion of workers with no wage change, while the blue bins, each with a width of 0.01, depict the range of wage changes.

Under the DNWR (asymmetric Calvo model), the wage setter's problem is

$$\begin{aligned}
& V(q_t(i), L_t, \frac{W_{t-1}(i)}{W_t}, x_t) \\
&= \max_{W_t(i)} \left[H(q_t(i), L_t, \frac{W_t(i)}{W_t}) \mathbb{I}(\frac{W_t(i)}{W_t} > \frac{W_{t-1}(i)}{W_t}) + \beta \mathbb{E}(V(q_{t+1}(i), L_{t+1}, \frac{W_t(i)}{W_{t+1}}, x_{t+1})) \right] \\
&+ \max_{W_t(i)} \left[H(q_t(i), L_t, \frac{W_t(i)}{W_t}) + \beta \mathbb{E}(V(q_{t+1}(i), L_{t+1}, \frac{W_t(i)}{W_{t+1}}, x_{t+1})) \right] \mathbb{I}(\frac{W_t(i)}{W_t} \leq \frac{W_{t-1}(i)}{W_t}) \mathbb{I}(x_t = 1) \\
&+ \left[H(q_t(i), L_t, \frac{W_{t-1}(i)}{W_t}) + \beta \mathbb{E}(V(q_{t+1}(i), L_{t+1}, \frac{W_{t-1}(i)}{W_{t+1}}, x_{t+1})) \right] \mathbb{I}(\frac{W_t(i)}{W_t} \leq \frac{W_{t-1}(i)}{W_t}) \mathbb{I}(x_t = 0).
\end{aligned}$$

If the current optimal wage is higher than the previous wage, wage setters can raise the nominal wages. However, if the current optimal wage is lower than the previous wage, wage setters can adjust downwardly only if x_t equals 1, with the probability of $(1 - \mu^{\text{DNWR}})$.

A.4.4 Solution Algorithm

This algorithm follows from [Heer and Maussner \(2009\)](#).

- Step 1: Guess a parameterized functional form of H and choose the initial parameter, γ_0 , γ_1 , and γ_2 .

$$\begin{aligned}
W_{t+1} &= H(W_t, M_{t+1}) \\
\ln(\frac{W_{t+1}}{W_t}) &= H(\ln(\frac{M_{t+1}}{W_t})) = \gamma_0 + \gamma_1 \ln \frac{M_{t+1}}{W_t} + \gamma_2 (\ln \frac{M_{t+1}}{W_t})^2
\end{aligned}$$

- Step 2: Solve the wage setter's optimization problem $V_t(q_t(i), L_t, \frac{W_{t-1}(i)}{W_t}, x_t)$, given the law of motion H .
- Step 3: Simulate the dynamics of the cross-sectional distribution for finite households for T periods using the policy function obtained by step 2.
- Step 4: Construct a time series for wage inflation. Burn first initial periods and estimate the parameters γ_0 , γ_1 , and γ_2 .

- Calculate simulated wage inflation, $\ln(\frac{W_{t+1}^S}{W_t})$,

$$\begin{aligned}\frac{W_{t+1}^S}{W_t} &= \frac{\left\{ \int \left[\frac{w_{t+1}(i)}{q_{t+1}(i)} \right]^{1-\theta} dj \right\}^{\frac{1}{1-\theta}}}{\left\{ \int \left[\frac{w_t(i)}{q_t(i)} \right]^{1-\theta} dj \right\}^{\frac{1}{1-\theta}}} \\ &\approx \left[\frac{\sum_j \left[\frac{w_{t+1}(i)/W_{t+1}}{q_{t+1}(i)} \right]^{1-\theta}}{\sum_j \left[\frac{w_t(i)/W_{t+1}}{q_t(i)} \right]^{1-\theta}} \right]^{\frac{1}{1-\theta}}\end{aligned}$$

- Estimate the parameters using the OLS

$$\ln\left(\frac{W_{t+1}^S}{W_t}\right) = H\left(\ln\left(\frac{M_{t+1}}{W_t}\right)\right) = \gamma_0 + \gamma_1 \ln \frac{M_{t+1}}{W_t} + \gamma_2 \left(\ln \frac{M_{t+1}}{W_t}\right)^2$$

- Step 5: Update γ_0 , γ_1 , and γ_2 using the OLS. Iterate from Step 2 to Step 5 until the parameters converge.
- Step 6: Test the goodness of fit for H using R^2 .

[Krusell and Smith \(1998\)](#) reported R^2 to check the accuracy of the predicted law of motion, and [Den Haan \(2010\)](#) argue that the maximum forecast error should be reported. R^2 is higher than 0.98,⁵ and the maximum forecast error is less than 0.1 percent.

A.4.5 The model implied cyclical of wage distributions

A.4.6 Symmetric menu cost model cost model

The menu cost model has two defining parameters: the probability of incurring a menu cost to adjust wages (μ^{Menu}), and the fixed cost (K). These parameters aim to match one specific moment - the average size of the spike at zero, leaving one free parameter. To evaluate the robustness of results from the Symmetric Menu cost models to parameter variations, Table [A25](#) demonstrates the cyclical properties of simulated wage change distributions as predicted by the model when μ^{Menu} varies from 0.4 to 1.⁶ The parameters are calibrated to align with an average spike at zero of 32.3%. The findings suggest that the Symmetric menu cost model consistently predicts greater responsiveness in the share of workers with wage cuts compared to the spike at zero, contradicting empirical evidence.

⁵ $R^{2,\text{Flex}} = 0.99$, $R^{2,\text{Calvo}} = 0.9$, $R^{2,\text{Menu}} = 0.99$, and $R^{2,\text{DNWR Menu}} = 0.98$, $R^{2,\text{DNWR Calvo}} = 0.97$.

⁶ There is no K less than 10 to match the spike at zero of 32.3% when $\mu^{\text{Menu}} \leq 0.3$.

Table A24: The model implied cyclicalities of wage distributions *with* measurement error (Frisch elasticity = 1)

	Spike at zero $\Delta W = 0$	Share of cuts $\Delta W < 0$	Share of raises $\Delta W > 0$	Spike at zero $\Delta W = 0$	Share of cuts $\Delta W < 0$	Share of raises $\Delta W > 0$
	A. Perfectly flexible			B. Calvo		
Employment	0.029 (0.031)	-0.306 (0.037)	0.277 (0.035)	0.056 (0.009)	-0.429 (0.018)	0.373 (0.019)
Inflation	-0.006 (0.010)	-2.996 (0.012)	3.002 (0.011)	-0.110 (0.011)	-2.463 (0.021)	2.574 (0.022)
	C. Long-term contracts			D. Symmetric menu costs		
Employment	-0.010 (0.013)	-0.277 (0.028)	0.286 (0.028)	-0.104 (0.017)	-0.267 (0.021)	0.372 (0.024)
Inflation	-0.013 (0.007)	-2.759 (0.017)	2.771 (0.016)	-0.840 (0.006)	-2.294 (0.008)	3.135 (0.009)
	E. DNWR (Asymmetric menu costs)			F. DNWR (Asymmetric Calvo)		
Employment	-0.514 (0.019)	-0.335 (0.013)	0.849 (0.024)	-0.595 (0.017)	-0.312 (0.012)	0.907 (0.022)
Inflation	-1.848 (0.018)	-1.372 (0.012)	3.220 (0.023)	-1.844 (0.018)	-1.345 (0.012)	3.189 (0.023)

The table reports the regression results using simulated data series with measurement error under six alternative wage-setting schemes, based on the equation (1), with Frisch elasticity being 1. Total 5000 periods of 5000 households are simulated. The numbers in parenthesis report standard error.

Table A25: The cyclical properties of wage change distributions predicted by Symmetric menu cost model by varying menu cost K and probability of facing menu cost μ^{Menu}

μ^{Menu}	K	The average Spike at zero (%)	The responsiveness to employment		
			(1) Spike at zero $\Delta W = 0$	(2) Fraction of $\Delta W < 0$	(3) Fraction of $\Delta W > 0$
1	0.0010	23.200	-0.120	-0.336	0.456
0.9	0.0012	23.035	-0.165	-0.333	0.498
0.8	0.0015	23.085	-0.187	-0.329	0.516
0.7	0.0020	23.205	-0.210	-0.358	0.568
0.6	0.0003	23.100	-0.210	-0.292	0.502
0.5	0.0004	23.000	-0.142	-0.353	0.495
0.4	0.0075	23.100	-0.164	-0.391	0.555
0.3	0.0190	23.164	-0.037	-0.469	0.506

Note: The table shows the responsiveness of the spike at zero, the share of workers with wage cuts and raises by varying parameters of Symmetric menu cost model, K and μ^{Menu} while targeting the spike at zero 32.3% without measurement error.